EVERGREEN ECONOMICS

1500 SW 1st Ave., Suite 1000
Portland, OR 97201
Office: 503.894.8676
www.evergreenecon.com

MEMORANDUM

Date: February 08, 2021

To: Mary Anderson, PG&E Codes and Standards Team

From: Evergreen Economics

Re: Air Purifier Study Results

This memo details initial findings from a survey that Evergreen Economics conducted for PG&E with the help of the National Opinion Research Center (NORC) at the University of Chicago. NORC recruited a representative sample of people who own air purifiers with a survey instrument developed by Evergreen. This memo details key findings from the survey about the following broad topics:

- Demographics of households that own purifiers;
- Air purifier types and incidence;
- Habits of households that own air purifiers and the settings they use; and
- Drivers of air purifier purchases.

Key Findings

Incidence and Purifier Characteristics

- Non-combination air purifiers (i.e., air purifiers that do not serve another purpose such as air conditioning or humidifying) have a nationwide incidence rate of 36 percent, but California residents are significantly more likely to own an air purifier, with an incidence rate of 48 percent.
- Nationwide, it is far more common for households to have one air purifier than to have multiple purifiers, though California households are more likely to report having multiple purifiers.
- Filters in air purifiers (i.e., HEPA and carbon filtration) are the most common type of purification method.
- Air purifiers that purify by multiple methods typically involve some combination of HEPA filtration in addition to another method.

- Households with two purifiers are more likely to have the same types of purifiers compared to those with more purifiers.
- Honeywell and Whirlpool are the most frequently encountered air purifier brands.
- Most reported air purifiers are three years old or newer, suggesting that the adoption of air purifiers is a relatively recent phenomenon. California residents were significantly more likely to have newer purifiers than non-California residents.
- Automatic fan speeds and Wi-Fi connectivity are not frequently encountered features. While the frequency of purifier types with Wi-Fi connectivity maps to the overall distribution of purifier types, less frequently encountered purifier types were more likely to have automatic settings (i.e., electrostatic and ozone generator purifiers).

Purchasing Motivations

- Combatting allergies and specific allergens such as dust and pollen were the strongest motivations for adopting air purifiers. Usually, respondents ranked both dust and pollen as reasons for adoption, suggesting allergies in general motivate adoption more than any one specific allergen.
- The reduction of smoke is a relatively infrequent motivator, though it was significantly more likely to be a motivator for California residents, with over 50 percent of California residents ranking smoke reduction as very or extremely important.
- Lower income respondents were more likely to report COVID-19 as a very or extremely important motivator for the adoption of air purifiers.
- The most important features of air purifiers to respondents were price, square footage cleaned, delivery rate, and noise level.

Purifier Usage

- Roughly half of the sample reported that their purifier usage did not differ by season, while the other half reported differences in usage.
- There were no strong trends by season in self-reported usage, though usage as reported by respondents dips slightly in winter months and increases slightly in summer months.
- Respondents in metro areas were more likely than those in non-metro areas to report changes in usage by quarter.
- Bedrooms were the most common areas where purifiers were run, and most respondents reported running their purifier in a dedicated room.
- Roughly half of respondents reported running their purifier less than six hours per day,

though nearly a quarter reported running their purifier all day long. Few respondents run their purifiers more than six hours a day if they are not consistently running the appliance.

- The daily reported usage frequency differs significantly by household size. Households with one or two members reported that 37 percent of their air purifiers are run all day, a much greater percentage than what is reported by larger households.

Notes About the Data

Because respondents reported owning multiple purifiers, many questions are presented at the purifier level, rather than at the household level. For example, when looking at how many purifiers belong to households with lower incomes, one lower-income household with three purifiers would count as three different observations, not as one. Exceptions to this (i.e., cases where reporting is done by household, not by purifier) will be noted where relevant.

This data was collected by AmeriSpeak, panel-based survey recruitment and distribution platform formed by the National Opinion Research Center at the University of Chicago. AmeriSpeak fielded a survey designed by Evergreen amongst a general population sample of US adults aged 18+ that also reported having at least one air purifier at home that exclusively purified the air and served no other purpose.

The sample was selected using 48 sampling strata, including strata based on age, race/ethnicity, education, and gender. Sample selection accounted for differential completion rates associated with different demographics to ensure that the selected respondents are a representative sample of the target population. In all, 5,929 people were invited to complete the survey, of whom 1,353 completed the screener, 503 were eligible, and 492 completed the survey.

In addition to the data collected by Evergreen, other data sources that may warrant future investigation are the Residential Appliance Saturation Survey (RASS)[1], which has evaluated the saturation of air purifiers in California in previous iterations. However, another major saturation survey, the Residential Energy Consumption Survey (RECS)[2] does not appear to have collected data on air purifiers in its past iterations.

[1] "2019 Residential Appliance Saturation Survey". California Energy Commission. Accessed from https://www.energy.ca.gov/data-reports/surveys/2019-residential-appliance-saturation-study

[2] "Residential Energy Consumption Survey". U.S. Energy Information Administration. Accessed from https://www.eia.gov/consumption/residential/index.php

Demographics

Table 1 outlines differences in demographic characteristics between those who qualified as eligible for our study (i.e., those with at least one air purifier that serves no other purpose) versus those who did not qualify as eligible (i.e., those without an air purifier and those with an air purifier that serves multiple purposes).

Overall, respondents in both eligibility groups had an average household size greater than the national average, and a much smaller proportion of them had a child in their household.

Table 1: Weighted Demographic Characteristics Qualified/Eligible Air Purifier Owners vs. Unqualified/Not Eligible Air Purifier Owners and Non-Owners

Eligibility Group	Average Household Size	Median Household Income	% with College Degree	% with Children in Household	% in the Western US	Median Age of Respondent
Eligible (n=491)	3.21	$60,000 - $74,000	35.08%	33.49%	26.82%	44 years
Not eligible (n=862)	2.80	$50,000 - $59,000	35.02%	25.34%	22.40%	48 years
All screened respondents (n=1,353)	2.95	$60,000 - $74,999	35.04%	28.25%	23.98%	47 years
National demographics	2.62	$62,843	32.1%	40%	22.52%	38 years

Note: National demographic information is from the United States Census Bureau.

Incidence and Types of Air Purifiers

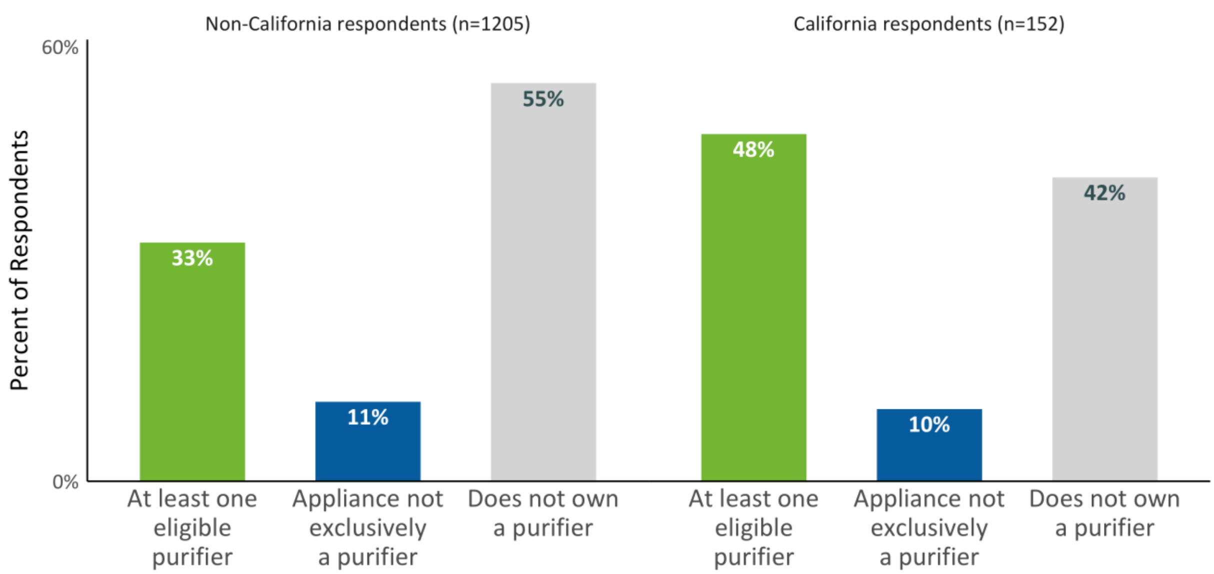
Incidence Rates

By looking at the percentage of sampled NORC respondents who reported owning an air purifier compared to those who did not, we can provide an estimate of the incidence rate of air purifiers in the US population.

Of the 1,353 panel members who completed the initial survey screener for eligibility, 640 reported owning an air purifier. However, this survey is focused on appliances that are exclusively air purifiers (as opposed to combined appliances such as dehumidifiers or air conditioners that also purify the air). Of those 640 respondents, 503 reported that some or all their purifiers do not serve any function besides purifying the air, and 492 of those respondents completed the survey. Thus, we estimate the current nationwide incidence rate of air purifiers to be **36 percent** for this study (the share of those with eligible purifiers relative to the number of those who completed the screener).

Notably, the incidence rate of air purifiers is higher in California than in the entire sample. While 33 percent of the non-California respondents reported having at least one eligible purifier, **48 percent of California** respondents **reported owning at least one eligible purifier** (Figure 1). This difference is statistically significant at 95 percent confidence.

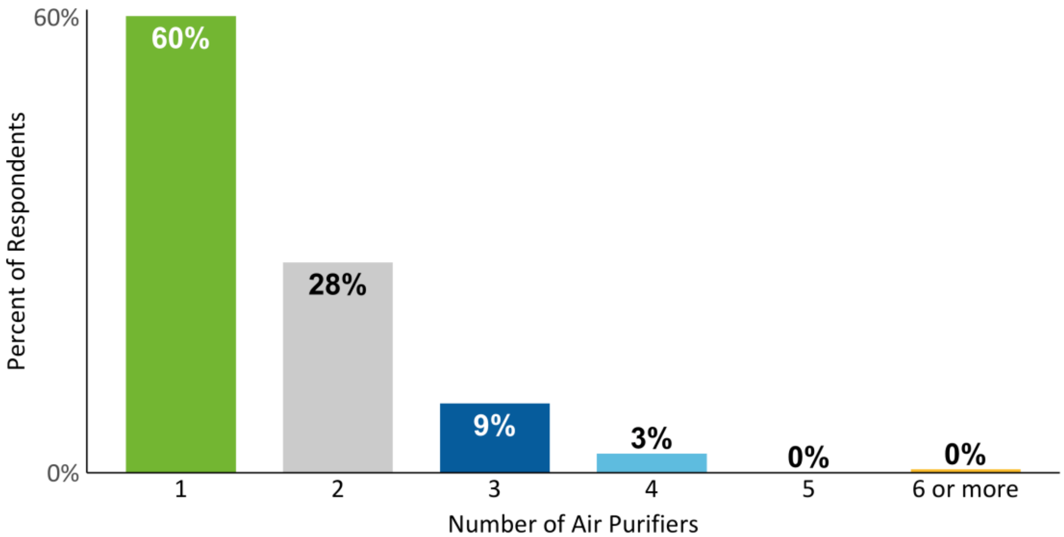
Figure 1: Incidence of Air Purifiers: California vs. Non-California Respondents



Number of Air Purifiers

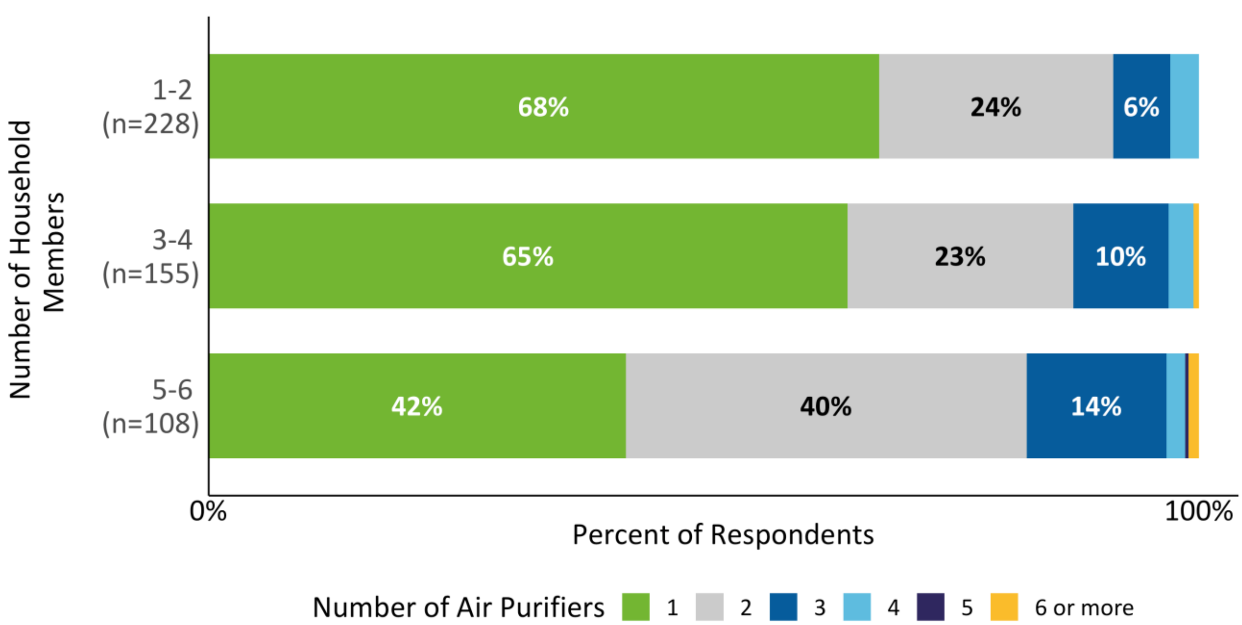
Of the 492 eligible respondents, most of them owned only one air purifier (60%) with the average number of reported purifiers being 1.56 air purifiers. Very few respondents reported owning more than four purifiers, with less than 1 percent of the sample reporting owning either five, six, or more purifiers (Figure 2).

Figure 2: Number of Air Purifiers per Household (n=492)



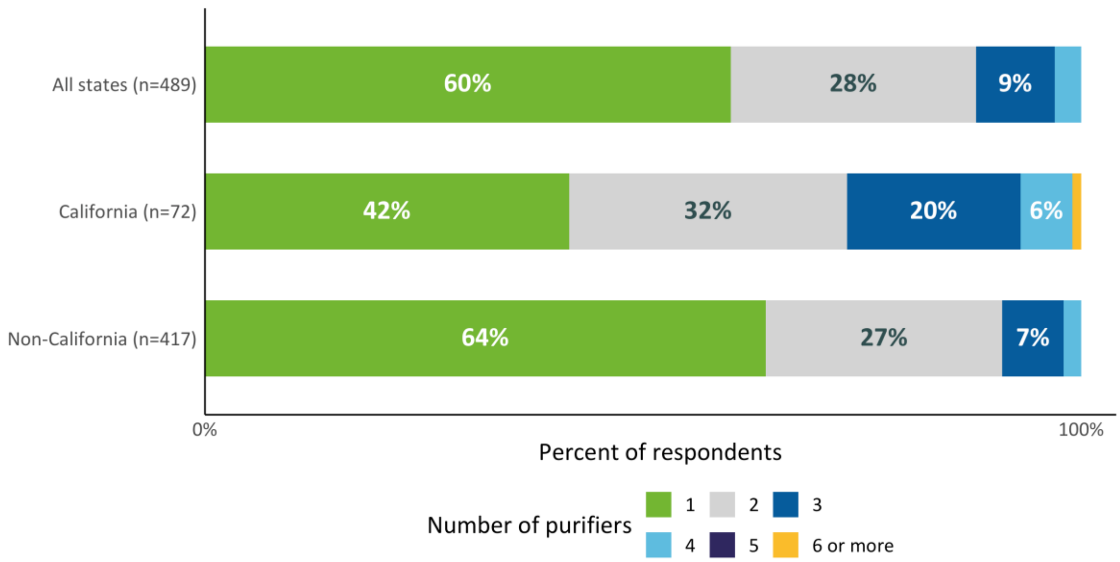
Households with fewer members tended to own fewer air purifiers (Figure 3). For households with one or two members, 32 percent own more than one air purifier, compared to 58 percent of households with five or six members. This difference is statistically significant at 95 percent confidence.

Figure 3: Number of Air Purifiers by Household Size



California residents were more likely to report owning more than one purifier than both the entire sample and respondents who live in states outside of California (Figure 4). Fifty-two percent of California residents own two or three purifiers, compared to 34 percent of respondents who live outside of California and 37 percent of the nationwide sample. The difference in reported numbers of purifiers between California residents and non-California residents is statistically significant at 95 percent confidence.

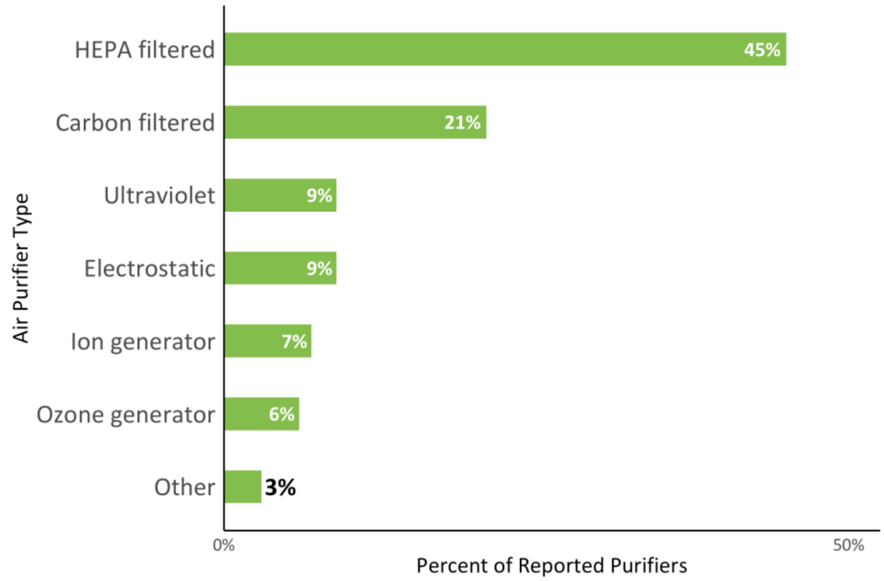
Figure 4: Number of Air Purifiers Nationwide and California v. Non-California



Types of Air Purifiers

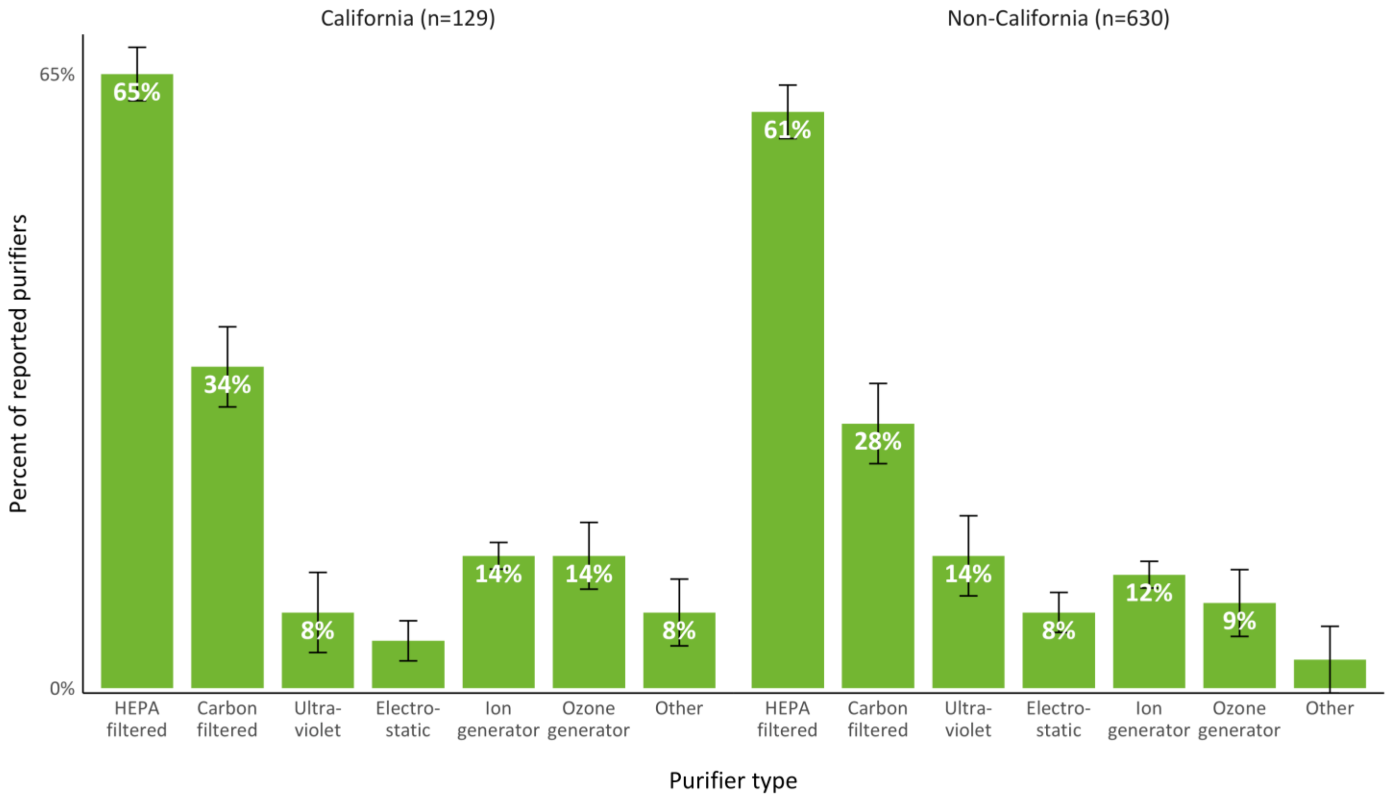
We asked respondents who reported owning at least one purifier to share the type of air purifier they own, and presented them with a range of options, including HEPA or carbon filtered, ion or ozone generating, and ultraviolet purifiers. Most respondents were able to report on the type of purifier they had at home, with this question only being skipped for seven of the reported air purifiers. Figure 5 summarizes the distribution of air purifier types across the sample.

Figure 5: Distribution of Air Purifier Types (n=764)



The distribution of air purifier types is roughly similar among residents of California and those not residing in California, with HEPA filtered air purifiers being most common among both groups (65% and 61%, respectively).

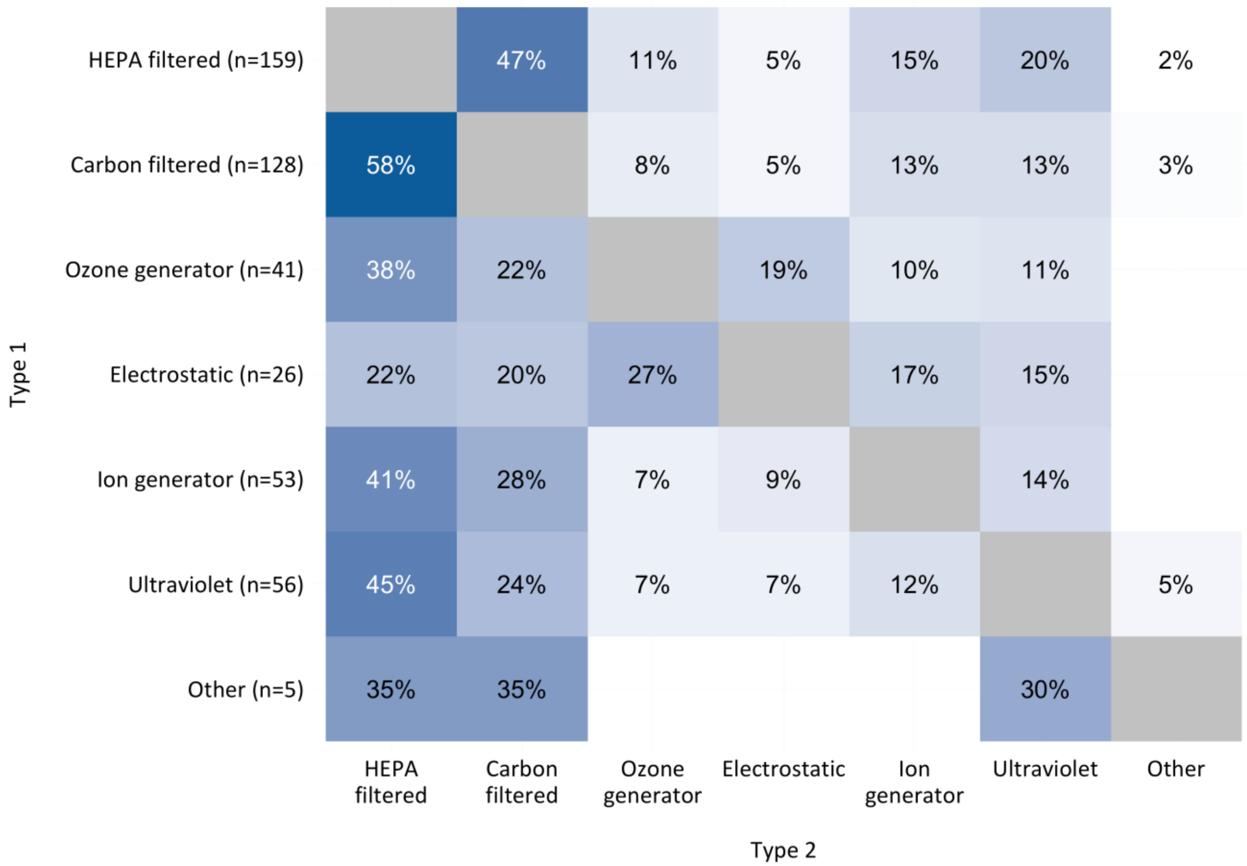
Figure 6: Purifier Types, California vs. Non-California



When presented with a range of types to choose from to describe their air purifiers, respondents were permitted to select more than one type for each purifier, as purifiers exist that purify the air using multiple methods (for example, using both HEPA filtration and UV light). To get a sense of the types of air purifiers that work using multiple methods, we calculated the percentage of purifiers of one type that were also of a different type (such as the percentage of carbon filtered purifiers that are also HEPA filtered purifiers). Figure 7 summarizes these results.

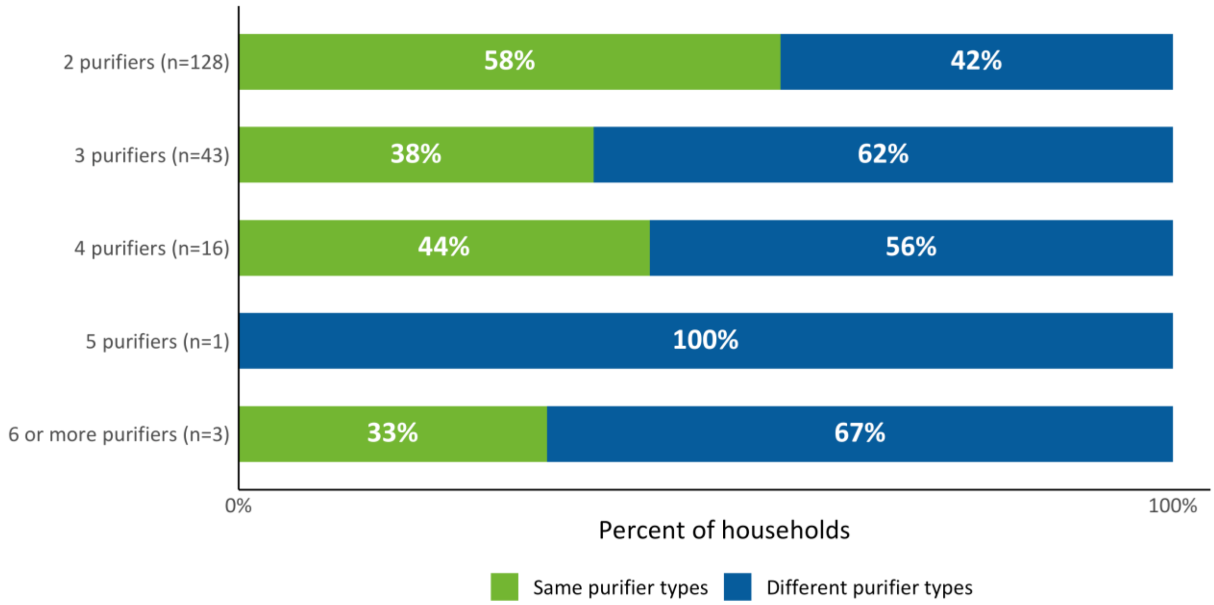
The most common combination was purifiers with both HEPA filtration and carbon filtration. Of reported purifiers with multiple types, 58 percent of carbon-filtering purifiers were also HEPA filtered. This was less likely to be the case for purifiers with HEPA filtration—only 47 percent of purifiers with HEPA filters also had carbon filtration. The other most common combinations for purifiers of multiple types were ion generators that also had HEPA filtration, and ultraviolet air purifiers that also had HEPA filtration.

Figure 7: Air Purifiers of Multiple Types



We also examined whether the air purifiers in households that reported having multiple purifiers were of same type of purifier or were different. Households with two air purifiers were much more likely to have the same type of air purifier than households with more than two air purifiers. Notably, 54 percent of purifiers owned by households with more than six air purifiers were the same type. On further investigation, households with two purifiers were statistically significantly more likely to have purifiers of the same type compared to households with three purifiers, though instances of households with higher numbers of purifiers were too infrequent for statistically significant differences to be detected.

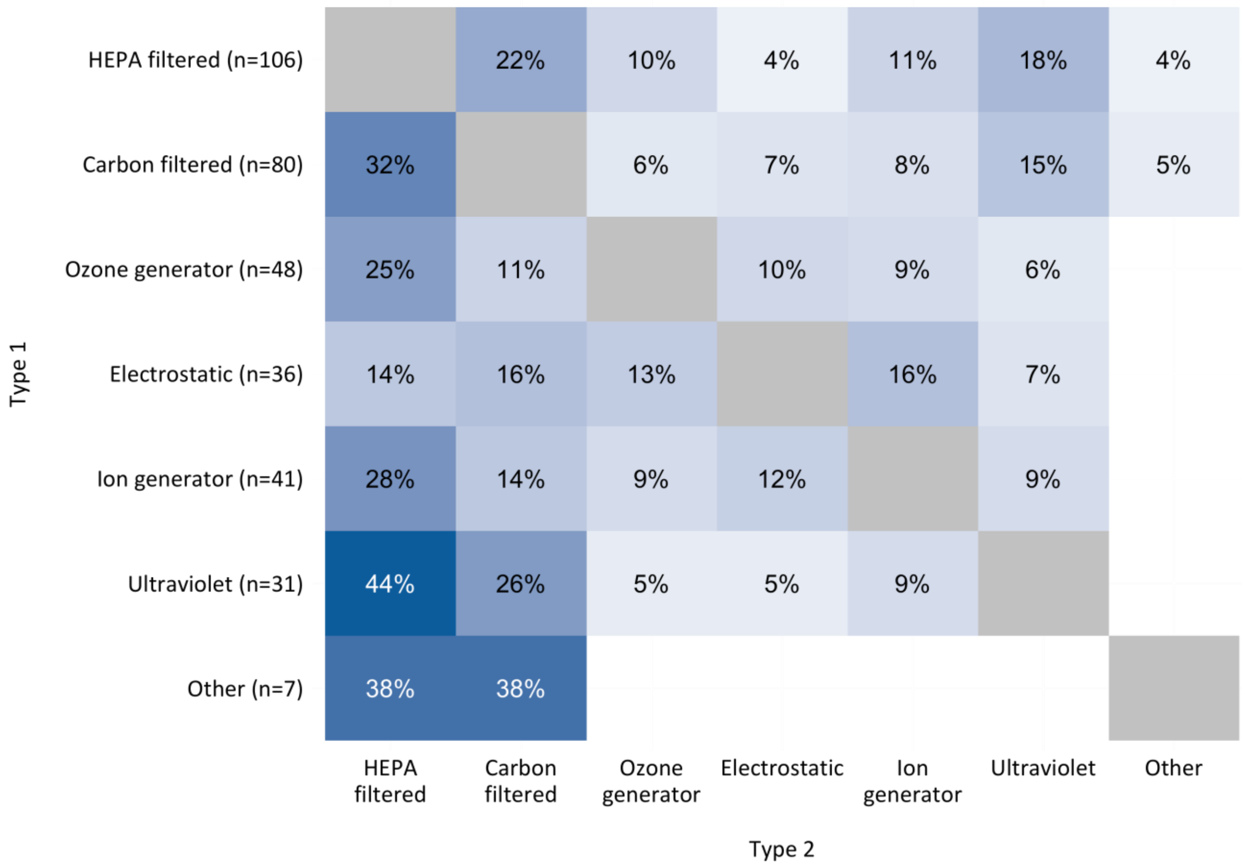
Figure 8: Share of Households with Multiple Purifiers of Different Types



For households with multiple purifiers of different types (e.g., ultraviolet, HEPA filtration), it would be useful to know how often these purifiers are paired together to determine the range of purifying options being implemented by households with a variety of purifiers. Figure 9 summarizes these findings.

There was a wide array of overlap between purifier types for households with multiple types of purifiers. The overlap between ultraviolet purifiers and purifiers with HEPA filtration is striking, with 44 percent of ultraviolet purifiers present in homes with HEPA filtration purifiers. It appears that based on this data, households that utilize numerous purifiers of different types are not trending strongly toward adopting any specific combination of types but implement a wide variety of combinations.

Figure 9: Most Common Pairings of Air Purifiers



Brands of Air Purifiers

The most common air purifier brand by far is Honeywell, which accounts for 29 percent of reported air purifiers. The more common air purifier brands are covered in Table 2, with Table 3 covering the less common air purifier brands. Both tables display the weighted shares of all reported purifiers that are of a certain brand, sorted from brands that accounted for the highest share of reported purifiers to those that accounted for the lowest shares. Only 3 percent of air purifiers are of an unknown brand.

Table 2: Most Common Air Purifier Brands (n=772)

Brand	Percentage of Reported Purifiers
Honeywell	29
Whirlpool	9
Philips	8
Dyson	6
Levoit	6
Sharp	4
Coway	3
GermGuardian	3
Winix	3
Blueair	2
Don’t know	3

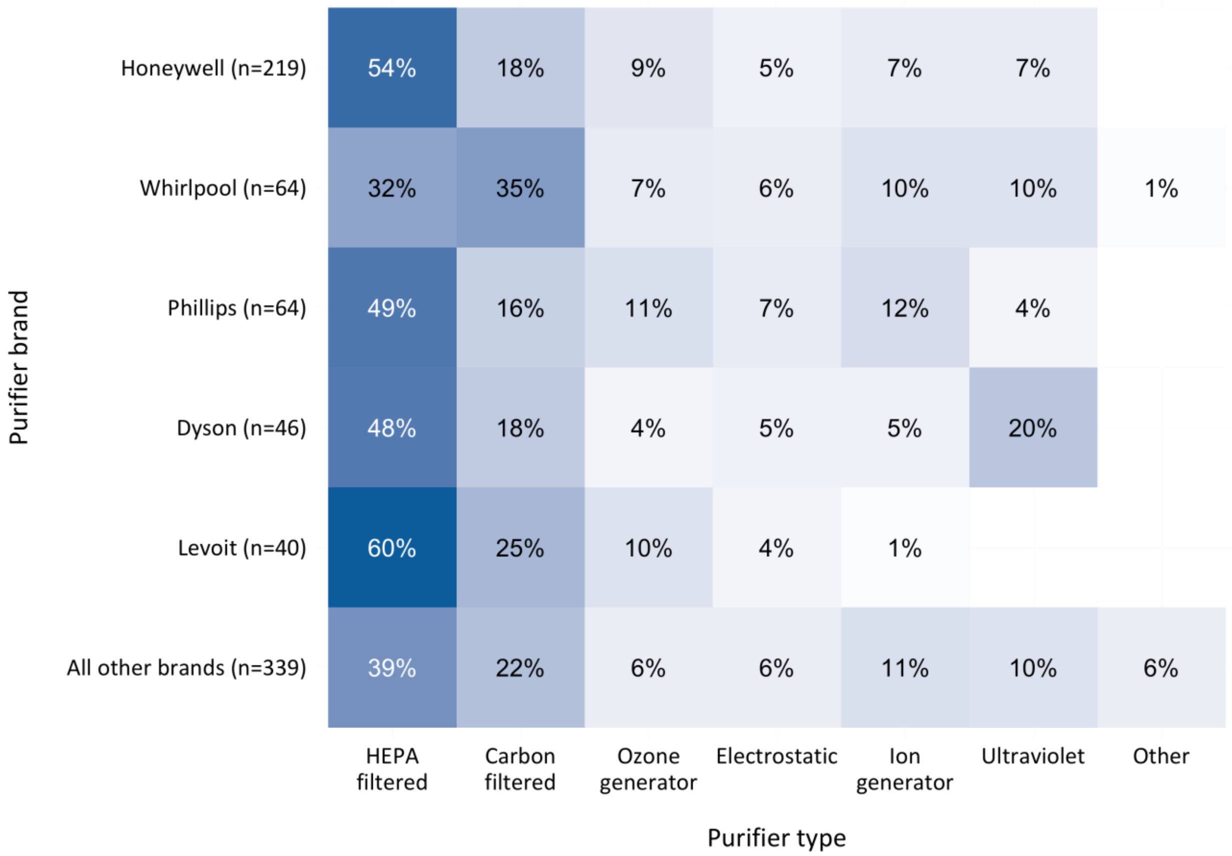
Table 3: Less Common Air Purifier Brands (n=772)

Brands			
Alen	IQAir	Airdoctor	Electrolux
Austin Air	Lennox	Air innovations	Fellowes
Ecoquest	Medify	Air Police	Filterqueen
Filtrete	Rabbit Air	Air Scrubber	Fresh Air
Holmes	Therapure	Airthereal	GPS
Homedics	TrueAir	Amazon Basics	Hampton Bay
Hunter	Newport	Atlas	Hauea
Ionic	Aer1	Bionaire	iDeer
Bissell	Crane	E-Z Breathe	Idylis
Coolmost	Critterzone	Eden Pure	Lentey
Kenmore	Koios	LG	LTLKY
Membrane Solutions	Oreck	Ozone	Partu
Prozone	Pure Enrichment	Pureair	Purely Products
Rabbit Air	Rainbow	Rowenta	Sharper Image
SilverOnyx	Storebary	Therapure	Trane
Trustech	Vanee	Vollara	Way Better
Westinghouse	Xiaomi	Zojirushi	

In addition to investigating the overall frequency of different air purifier brands, we investigated the degree of overlap between the most common brands and the types of air purifiers reported by respondents. These results are summarized in Figure 10.

For most brands, nearly half or more than half of reported purifiers were HEPA filtered, except for Whirlpool purifiers, a third of which were HEPA filtered and another third of which were carbon filtered (Figure 10). While a high proportion of Levoit purifiers were HEPA filtered, they were reported to be the second most likely brand of purifier to be carbon filtered, behind Whirlpool purifiers. These results continue to show the dominance of HEPA filtered products, regardless of the brand that consumers choose to purchase. This fact was true of all other purifier brands besides the top five most frequently reported brands. Subsequent analysis of purchasing data can test the hypothesis supported by the present data that across brands, HEPA filtered purifiers are the most popular.

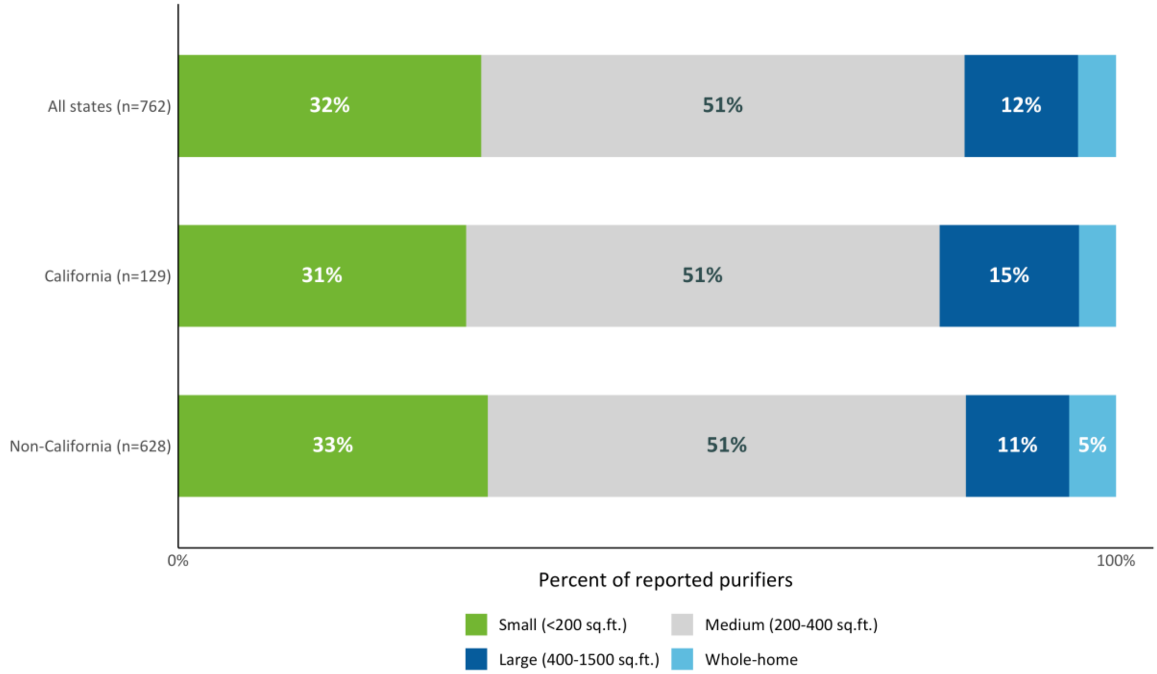
Figure 10: Most Common Combinations of Purifier Brands and Types



Sizes of Air Purifiers

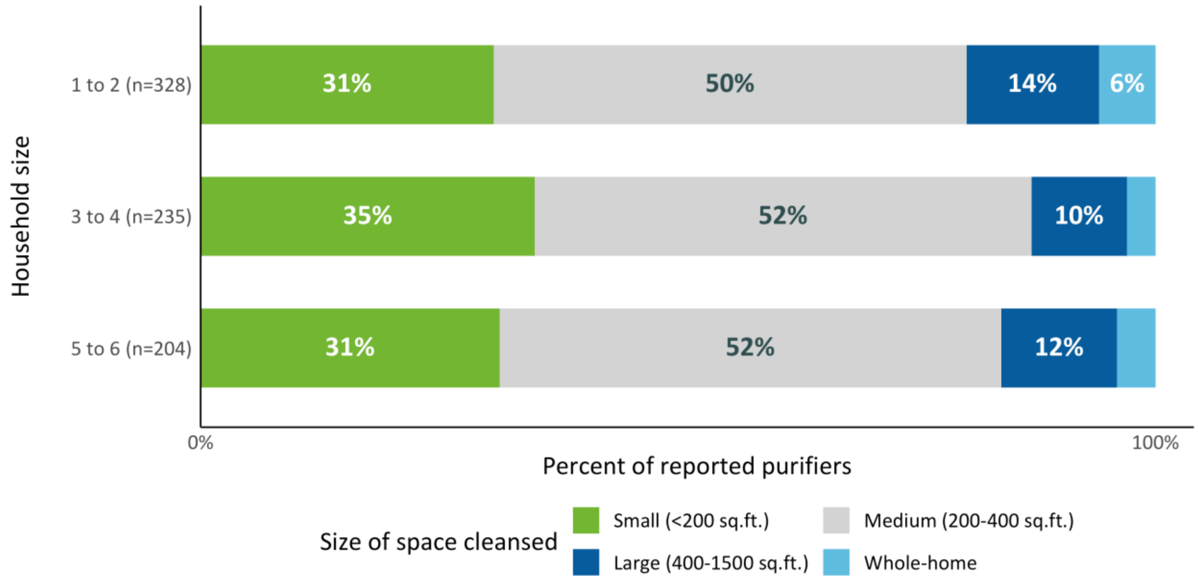
Respondents were asked about the amount of space their air purifiers can filter by square footage. The overall distribution of air purifiers by size filtered does not differ much between those who own air purifiers and reside in California and those who reside in other states. Regardless of state of residence, the most common space that air purifiers can cleanse is a medium size of 200-400 square feet with this being the reported size for 51 percent of air purifiers (Figure 11).

Figure 11: Overall Distribution of Air Purifier Sizes



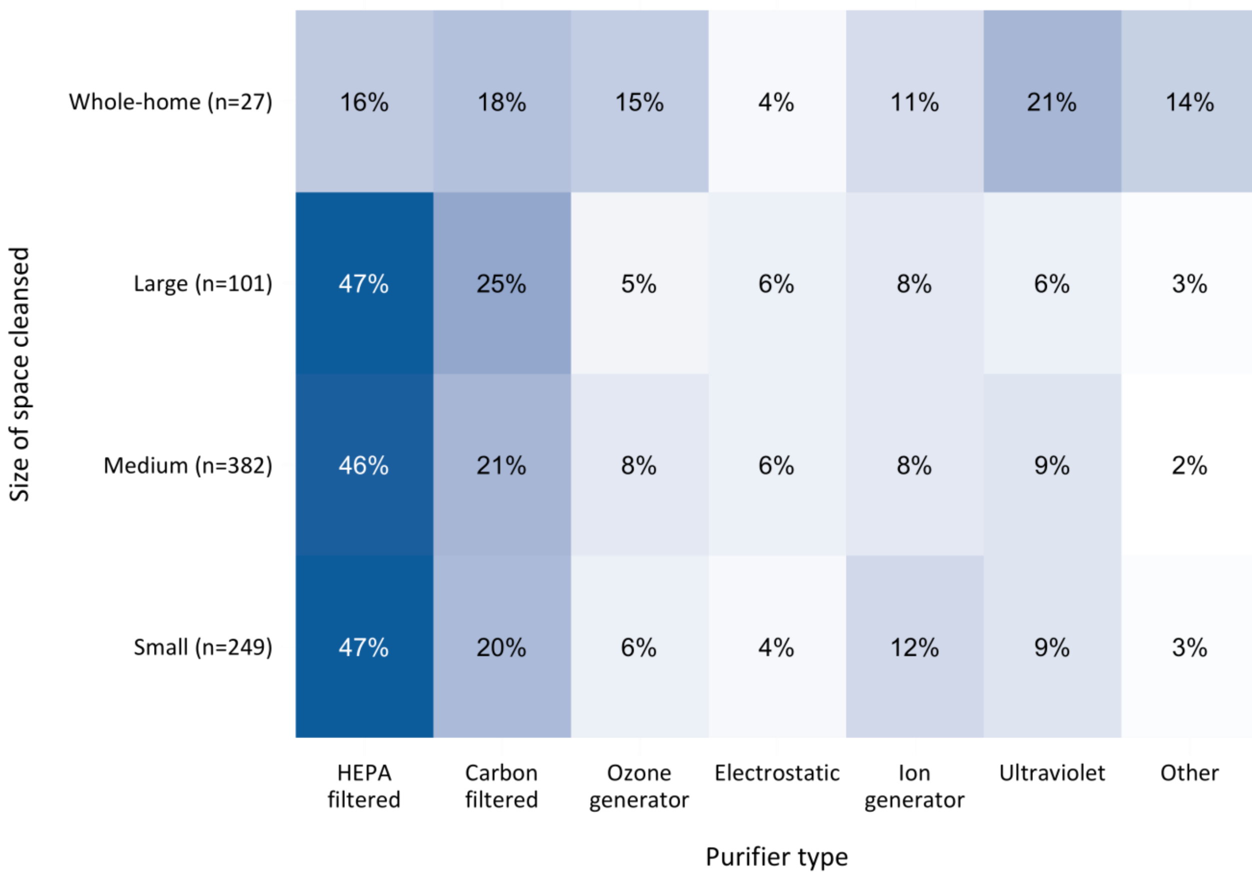
The size space that purifiers can cleanse did not appear to vary substantially by respondents' household size. Smaller households (i.e., those with one or two residents) were more likely to report large or whole-home purifiers than medium-sized or larger households (Figure 12). Households with three or four residents were more likely to report owning small air purifiers than smaller or larger households.

Figure 12: Size of Space Cleansed by Household Size



For purifiers of most sizes, the distribution of air purifiers followed the distribution of air purifier types across the entire pool of purifiers, with the majority of most being either HEPA filtered or carbon filtered. There is no clear trend for air purifier types for whole-home purifiers—proportions were similar for most air purifier types for whole-home purifiers, except for electrostatic whole-home purifiers, which were relatively less frequent (Figure 13).

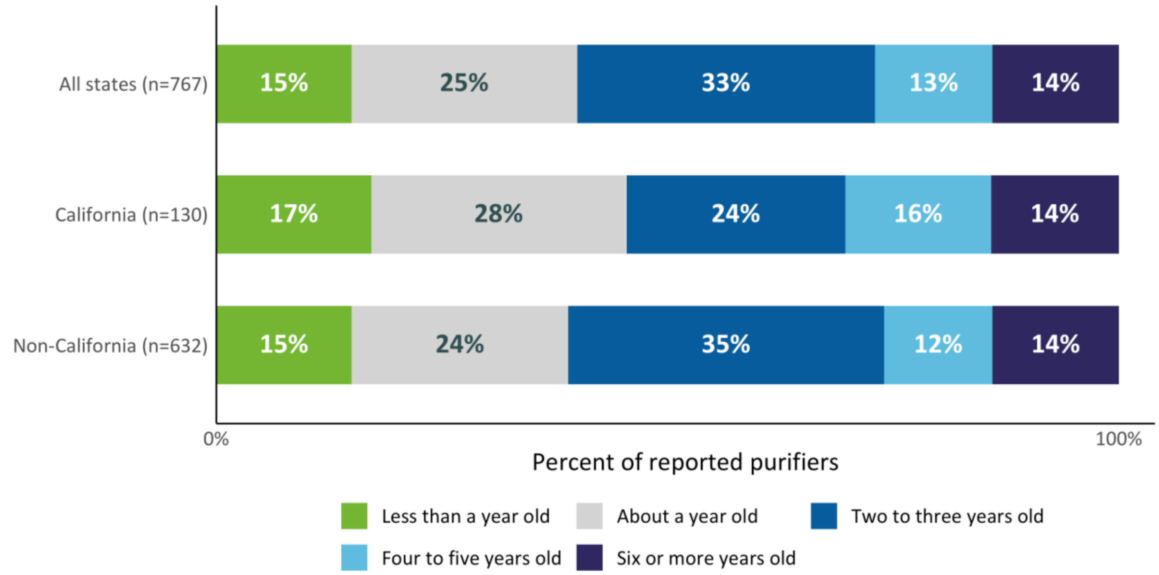
Figure 13: Size of Space Cleansed by Type



Age of Air Purifiers

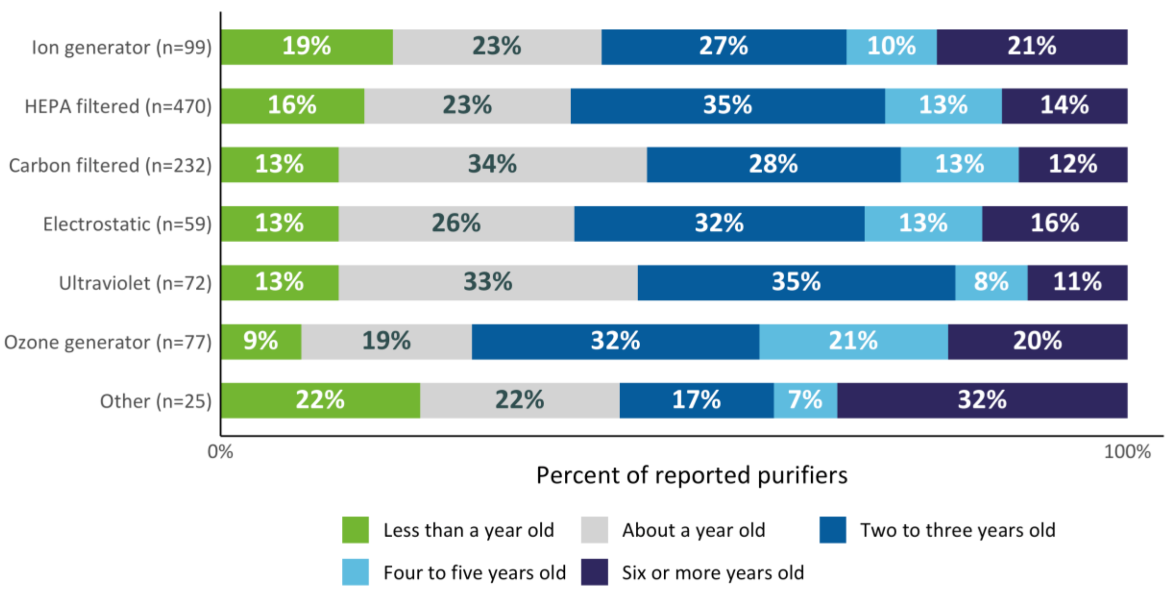
The age of respondents' purifiers varied, but most purifiers were reported to be three years old or newer, suggesting that the adoption of air purifiers is relatively recent for most households (**Error! Reference source not found.**). The age of the reported purifiers differed for California residents and non-California residents. Forty-three percent of purifiers in California were reported to be a year old or newer, compared to 29 percent of purifiers outside of California and 40 percent nationwide. This difference between California and other states is statistically significant at 95 percent confidence, suggesting that California residents are significantly more likely to have bought a purifier recently compared to other states.

Figure 14: Age Distribution of Air Purifiers



Ozone generator, ion generator, and electrostatic purifiers were more likely to be older than other types of purifiers (i.e., four years or older, Figure 15). Across all types of purifiers, the majority are three years old or newer, suggesting that no matter the type of purifier that consumers choose, their decision to purchase one is a relatively recent occurrence.

Figure 15: Purifier Type by Age

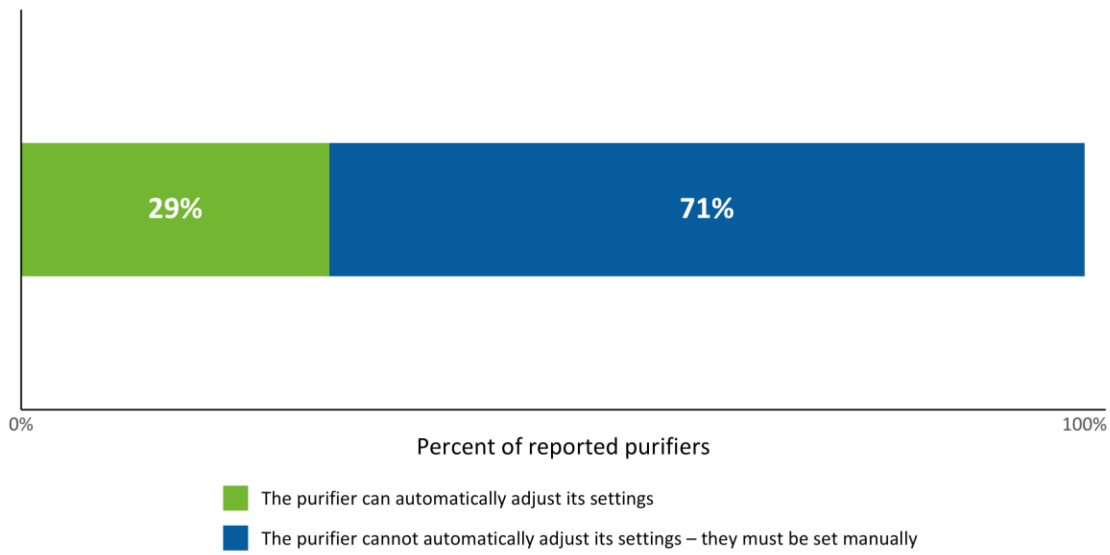


Air Purifier Settings

Automatic Settings

Among the questions respondents were asked regarding their air purifier settings was whether a given purifier was capable of automatically adjusting its settings based on ambient temperature. Based on respondent self-reporting, the incidence of purifiers with automatic settings is low, with only 29 percent of reported purifiers including this feature (Figure 18).

Figure 16: Incidence of Purifiers with Automatic Settings (n=764)



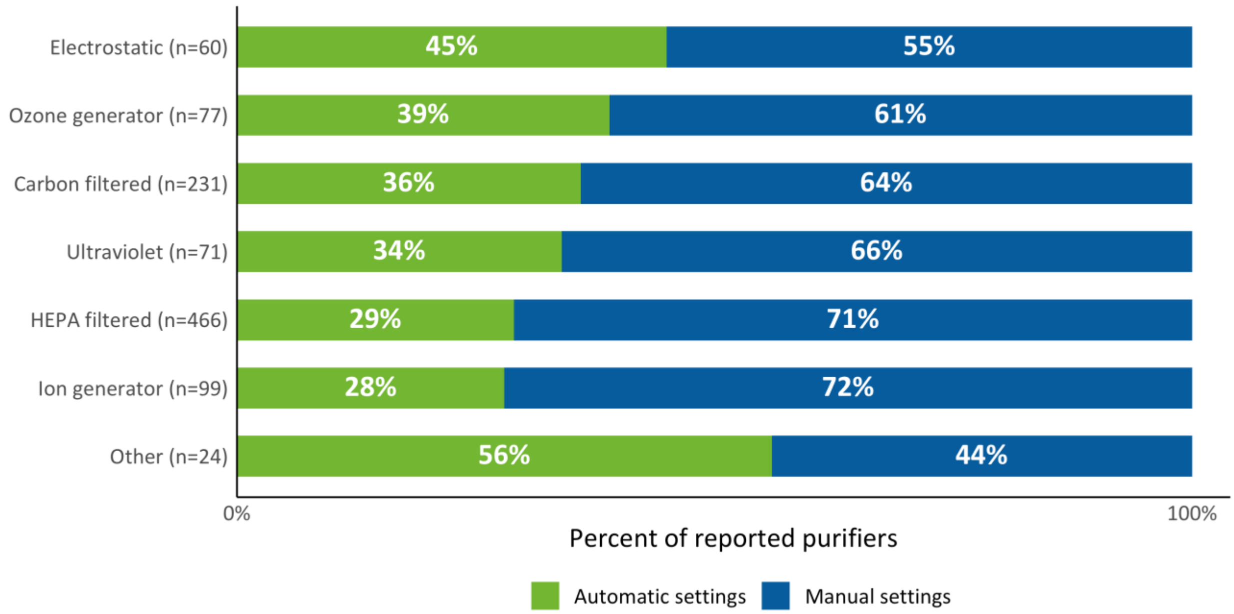
The most common brands of purifiers with automatic settings overlapped to a degree with the most common purifier brands across all purifiers (i.e., Honeywell and Philips being the two top reported brands; see Table 4). However, some brands were more likely to appear for automatic purifiers than for the overall pool of purifiers, such as Dyson and Winix.

Table 4: Most Common Brands of Purifiers with Automatic Settings

Brand	Percentage of Purifiers with Automatic Settings (n=226)	Percentage of All Purifiers (n=772)
Honeywell	28	29
Philips	12	8
Dyson	11	6
Whirlpool	10	9
Winix	8	3
Coway	5	3
Alen	3	1
Levoit	3	6
Sharp	3	4

The distribution of air purifier types with automatic settings differs from the overall distribution of air purifiers. While purifiers with HEPA or carbon filtration were the most frequently encountered types across the entire sample, electrostatic and ozone generator purifiers were the most likely to have been reported to have automatic settings (Figure 17). Given that these are not the most prevalent types of purifiers overall, this raises the question of how many HEPA and carbon filtration purifiers are available with automatic settings, and whether those sell less than their standard equivalents. Future analysis of purifier sales data could help provide answers to this question.

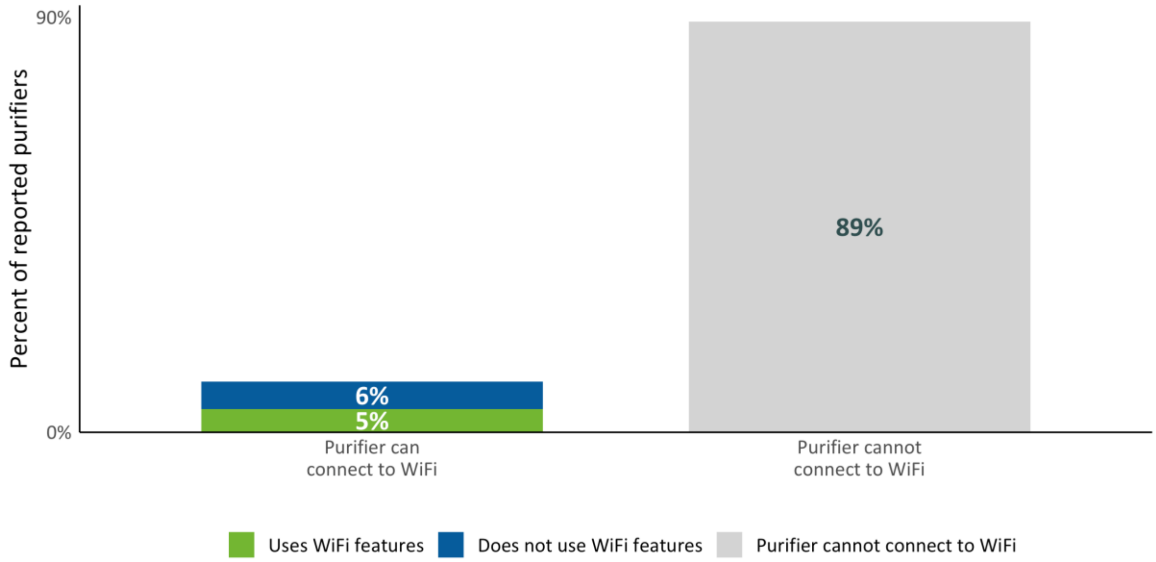
Figure 17: Distribution of Types of Air Purifiers with Automatic Settings



Wi-Fi Connectivity

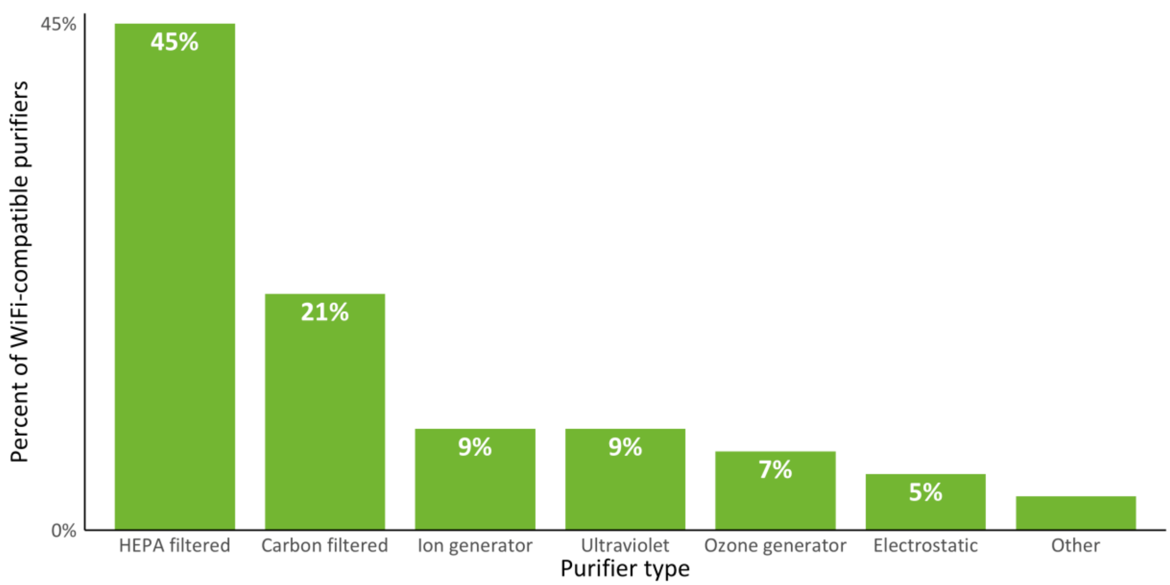
Another question asked of respondents was whether their purifier could connect to Wi-Fi. Overall, it was very uncommon for purifiers to be Wi-Fi compatible. Of the 11 percent of the purifiers that can connect to Wi-Fi, 6 percent of them are actively connected and use WiFi features, while 5 percent of them do not. (Figure 18). This suggests that smart purifiers are far from common across the nation, and even adopters rarely use the Wi-Fi-compatible features.

Figure 18: Percentage of Purifiers that are Wi-Fi Compatible (n=765)



Unlike with automatic fan speed, the types of purifiers that are Wi-Fi compatible more or less match the distribution of types seen across the entire sample of reported purifiers (Figure 25). HEPA and carbon filtration purifiers are the most likely to be Wi-Fi compatible, with all other types comprising less than half of Wi-Fi compatible purifiers.

Figure 19: Types of Wi-Fi-Compatible Air Purifiers (n=97)



Air Purifier Purchasing Motivations

Survey respondents were presented with a series of reasons for choosing to purchase air purifiers for their homes and were asked to rate the importance of those reasons from not at all important to very important. Figure 20 summarizes the distribution of responses across all possible purchasing motivations. Note that purchasing motivations were asked across all reported purifiers, so we present results here in terms of respondents, not in terms of reported purifiers.

In general, it seems that the purchase of air purifiers is more strongly motivated by combating allergies and specific allergens. 88 percent of respondents highlighted allergies as an extremely, very, or moderately important motivator while 86 percent respondents rated dust and 85 percent rated pollen as an important motivator (very, extremely, or moderately). Reducing germs was also highly ranked but less so than allergens, with 76 percent rating germ reduction as important to some degree.

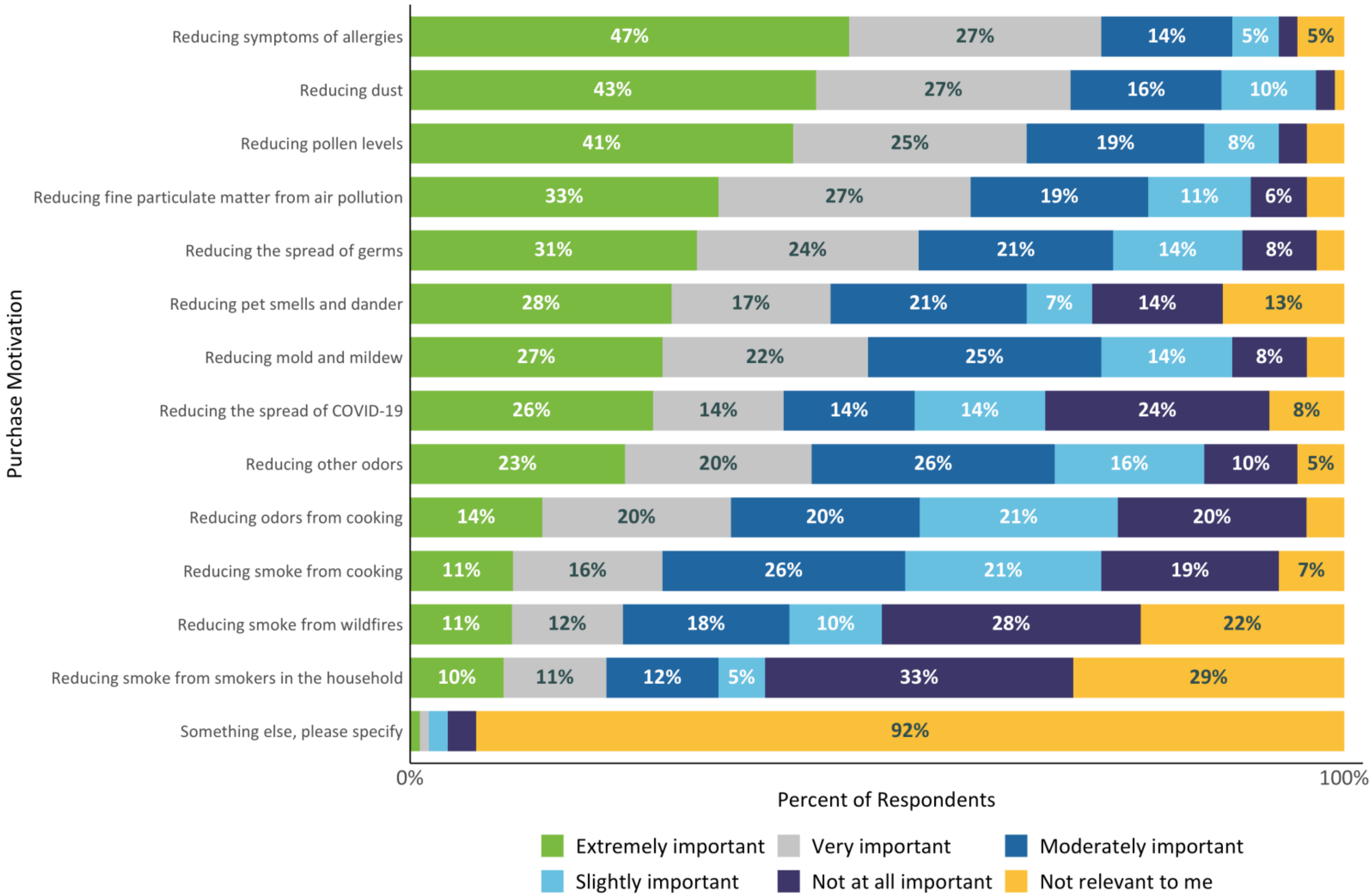
Smoke reduction is not as salient a factor across the whole sample, with 41 percent reporting wildfire smoke as extremely, very, or moderately important reasons for their purchase of air purifiers, and only 33 percent saying the same of smoke from smokers in the household. In fact, reducing smoke from household smokers was the factor most frequently rated as not at all important, and wildfire smoke was the second most frequently rated as not at all important.

Odor reduction is also a less salient motivator for purchasing air purifiers than reduction in allergies, with 69 percent of respondents ranking reducing odors in general as very, extremely, or moderately important and 54 percent of respondents rating reducing cooking odors as a extremely, very, or moderately important purchase motivator.

In summary, respondents' motivators for purchasing air purifiers correspond to the following categories, in order of importance:

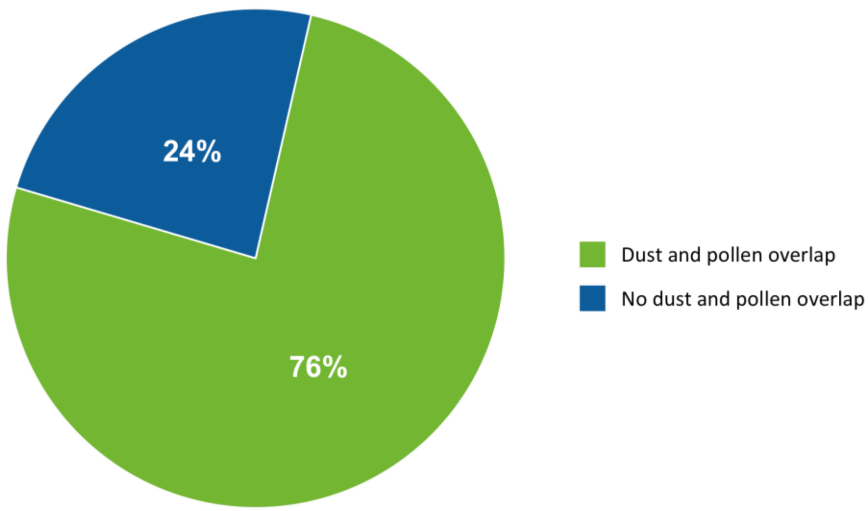
- Dust;
- Pollen;
- Particulates/Pollution;
- Biological agents (i.e., mold and germs excluding COVID-19);
- Odors; and
- Smoke (both wildfire and from smokers at home)

Figure 20: Overall Air Purifier Purchasing Motivations (n=491)



Given that allergies appear to be a strong motivator for the adoption of air purifiers, we investigated the degree to which the importance of reducing different allergens overlap for respondents, specifically for pollen versus dust allergies (Figure 21). In over half of the cases where these two items were listed as important to a degree, they were selected together. This points further to the fact that allergies collectively, not just specific allergies, are a motivator for purchasing air purifiers.

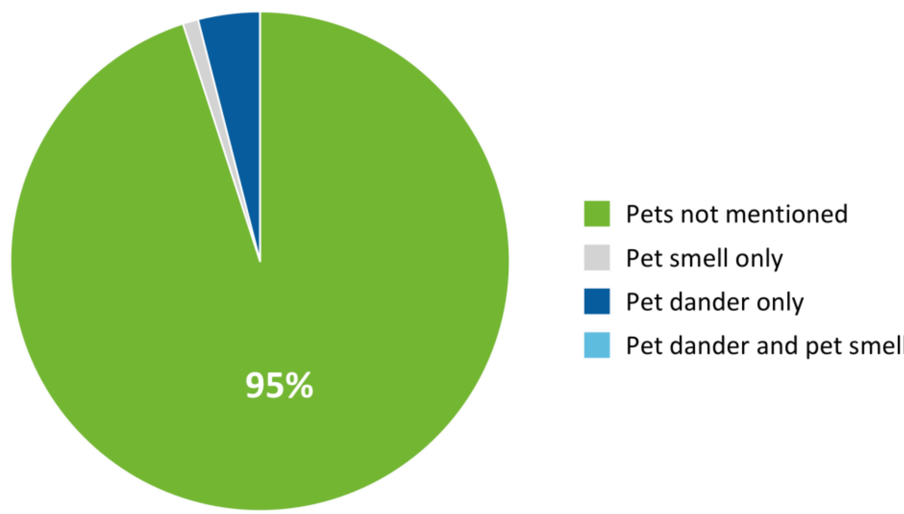
Figure 21: Importance of Pollen Versus Dust in Adoption of Air Purifiers



In our close-ended questions regarding purchasing motivations, we asked respondents the degree to which pet dander and/or pet odors contributed to their purchase of an air purifier. By asking about these motivators together, it is difficult to get a picture of the degree to which either pet smells, pet dander, or both contributed to respondents choosing to buy an air purifier.

To get a clearer view of this, we investigated open-ended responses to questions regarding purchase motivations for those who specifically mentioned pet hair or pet dander. These issues were not top of mind for most respondents (Figure 22). Only 7 percent of the overall sample reported concerns related to pets as a motivator in response to the open-ended question. Of those who did, pet dander specifically was the greatest motivator, followed by pet smells. Very few respondents reported a combination of pet dander and smells as the reason for their adoption of an air purifier.

Figure 22: Pet Dander vs. Pet Smells as a Purchase Motivator (n=491)



Though wildfire smoke was not a very salient purchasing motivator across the entire sample, it was more salient for respondents living in California that for those outside of California; 77 percent of California respondents rated wildfire smoke as very, extremely, or moderately important, compared to 34 percent of non-California respondents (**Error! Reference source not found.**23 and Figure 24).

This difference was found to be statistically significant at 95 percent confidence. The importance of pollution and particulates also tended to be of more importance to California respondents, 86 percent of whom rated it as very, extremely, or moderately important, compared to 79 percent of those outside California cases. In contrast, germs were a slightly stronger motivator for those outside California (78 percent rated extremely, very, or moderately important) than those in California (68%). In most other respects, data from California respondents about purchasing motivations mapped fairly closely to non-California respondents.

In summary, the importance of different motivators are largely the same for those within California and those outside California, with the exception of wildfire smoke, which as important to respondents in California as biological agents.

Figure 23: Purchase Motivations for California Respondents (n=72)

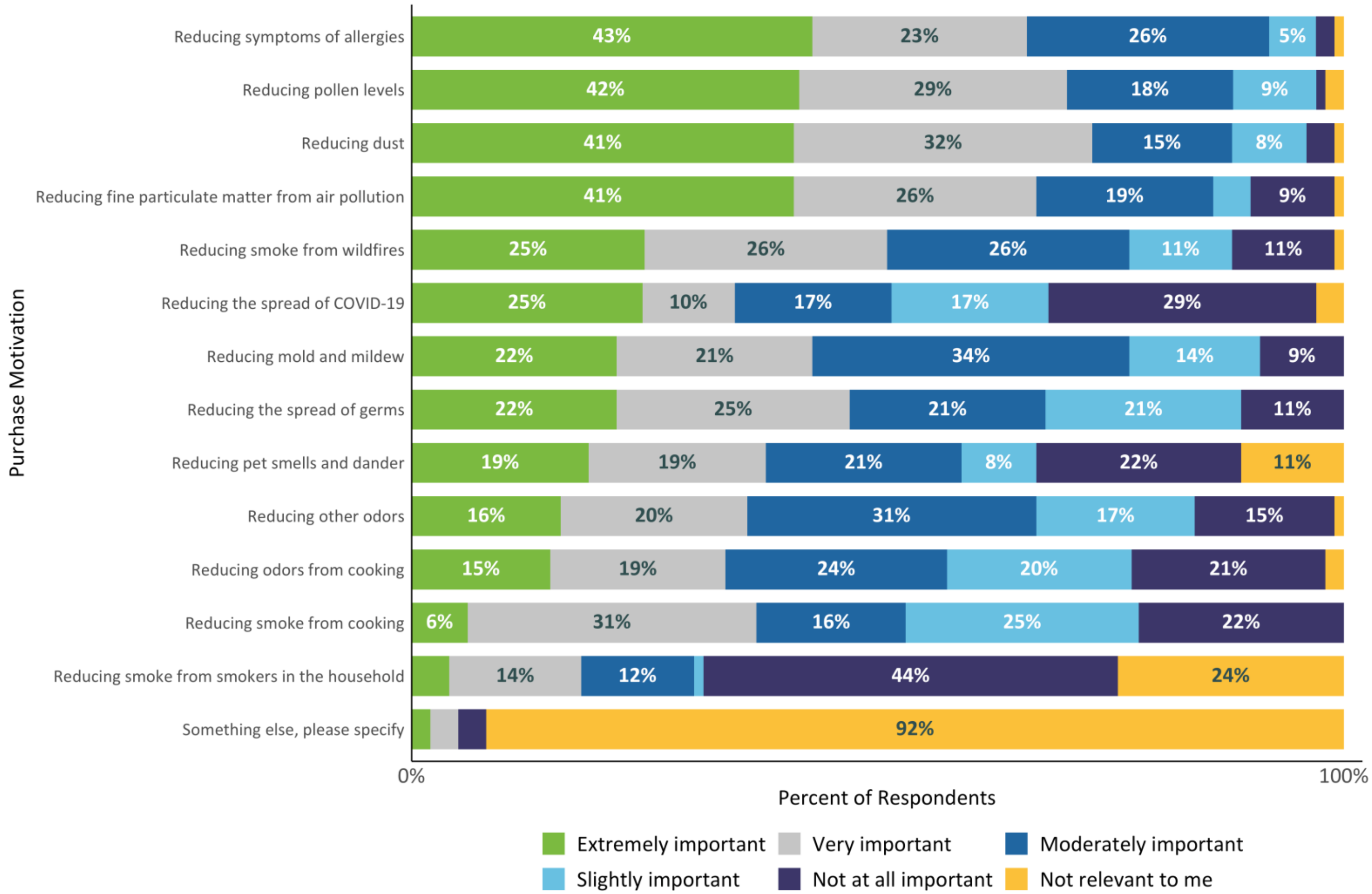
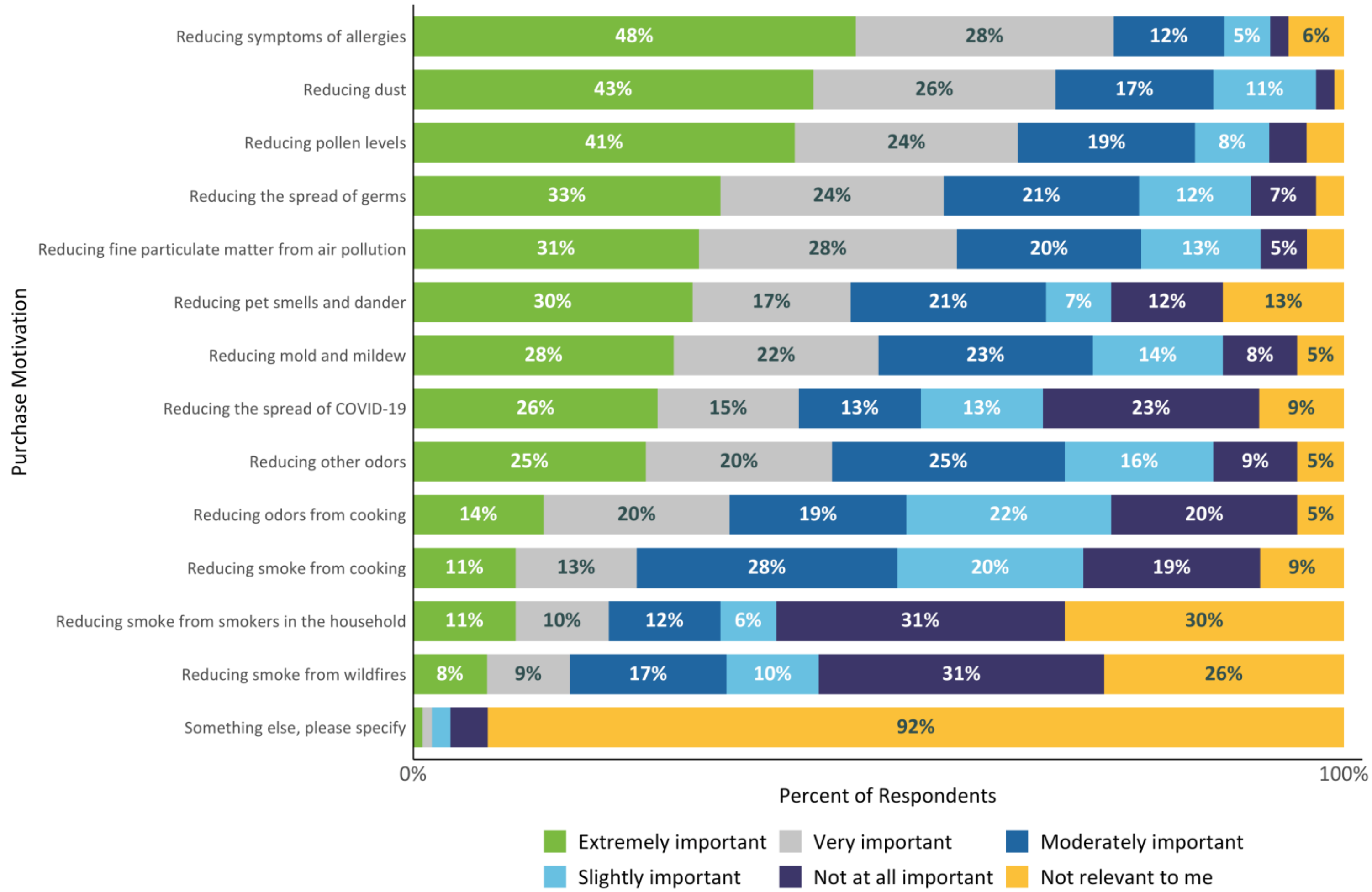
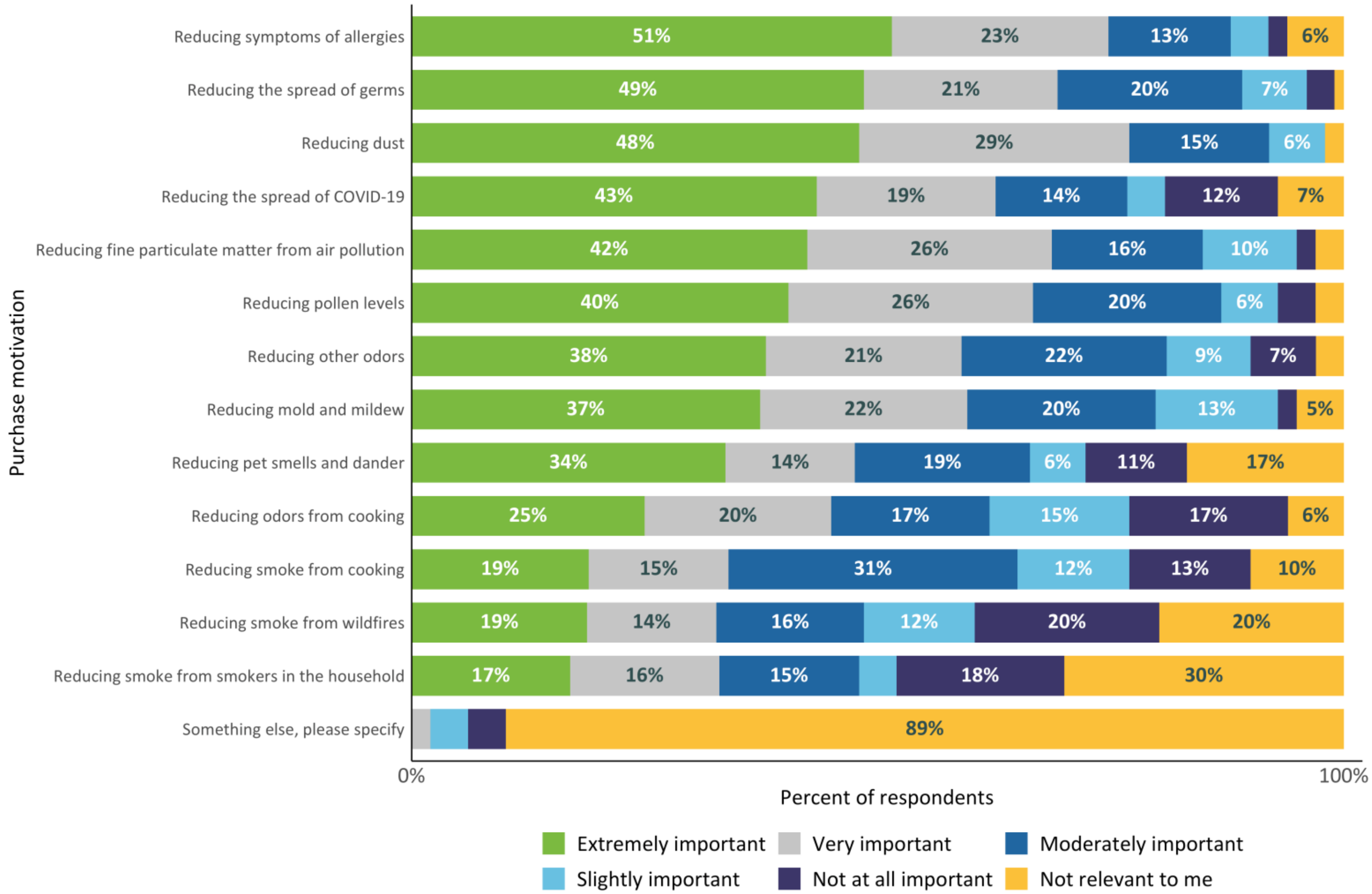


Figure 24: Purchase Motivations for Non-California Respondents (n=419)



Those with lower household incomes (under $30,000 dollars per year) were as likely if not more likely to rank allergy-related reasons as strong motivators for purchasing air purifiers; however, there are other points of divergence. For example, while only 40 percent of the entire sample reported reducing the spread of COVID-19 as an extremely or very important motivation for their purchase of an air purifier, 62 percent of lower-income respondents reported that COVID-19 was a very or extremely important motivator (Figure 25). Similarly, lower-income respondents were more likely to report reducing the spread of germs as a motivator for their purchase, with 70 percent reporting that this was an extremely or very important motivator, compared to 55 percent of the entire sample.

Figure 25: Purchase Motivations for Lower-Income Households (n=113)



When comparing respondents over the age of 65 and respondents under the age of 65, we found that seniors more often rated the need to reduce fine particulate matter from air pollution as extremely or very important more often than non-seniors (68% vs. 58%). Another difference arose when we compared respondents with and without a child at home. Respondents without a child at home were more likely to say that reducing dust was extremely or very important in their decision to purchase an air purifier when compared to households with a child (73% vs. 65%).

Finally, when comparing responses between those living in metro areas and non-metro areas, we found that respondents in non-metro areas were more likely to respond that reducing dust, pet smell and dander, symptoms and causes of allergies, odors from cooking, and other odors were extremely or very important in their decisions, compared to those living in metro areas (Table 5).

Table 5: Purchase Factors Reported as Extremely or Very Important by Metro and Non-Metro Areas

Factor	Metro Areas	Non-Metro Areas
Reducing dust	69%	74%
Reducing pet dander or odor	44%	53%
Reducing symptoms of allergies	73%	80%
Reducing odors from cooking	32%	45%
Reducing other odors	42%	55%

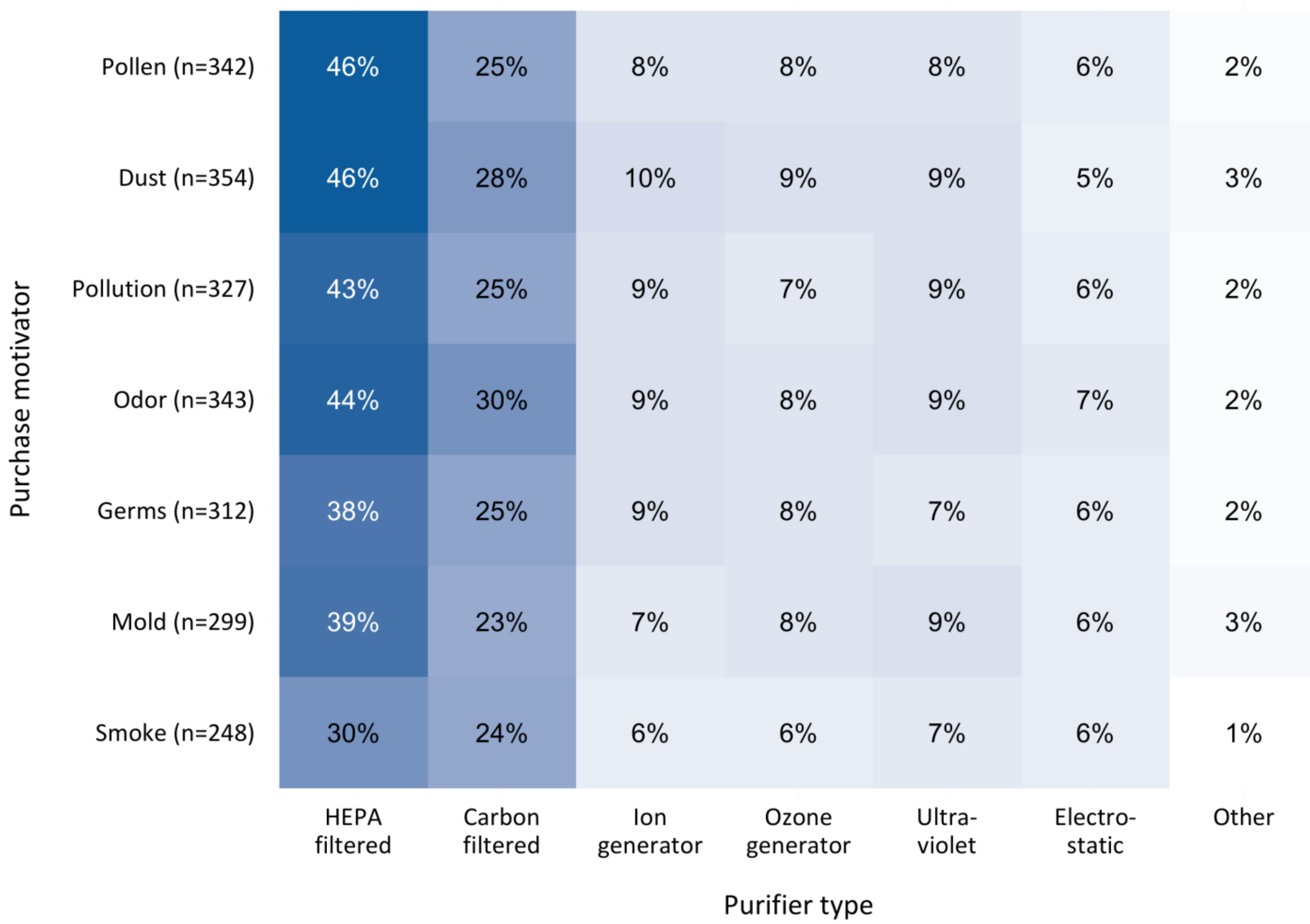
Purchasing Motivations and Purifier Types

To determine the extent to which the aforementioned purchasing motivations translate into specific decionmaking about which purifier to buy, we examined the distribution of purifier types for those who described different motivators as more or less important.

Specifically, Figure 26 below shows the distribution of purifier types among those who described pollen, dust, particulates/pollution, biological agents, odors, and/or smoke as important to a degree (i.e., extremely, very, or moderately important). This analysis is presented on the *household* level, rather than the purifier level. For example, 83 percent of households that self-reported pollen as a purchase motivator also reported owning at least one HEPA filter.

Overall, HEPA filters are the most popular choice regardless of the purchasing motivation considered. Less common purifier types, such as ion and ozone generators, tended to be most popular for reducing dust, but differences between any given purchasing motivator are small.

Figure 26: Purifier Types by Major Purchase Motivators



Importance of Purifier Characteristics

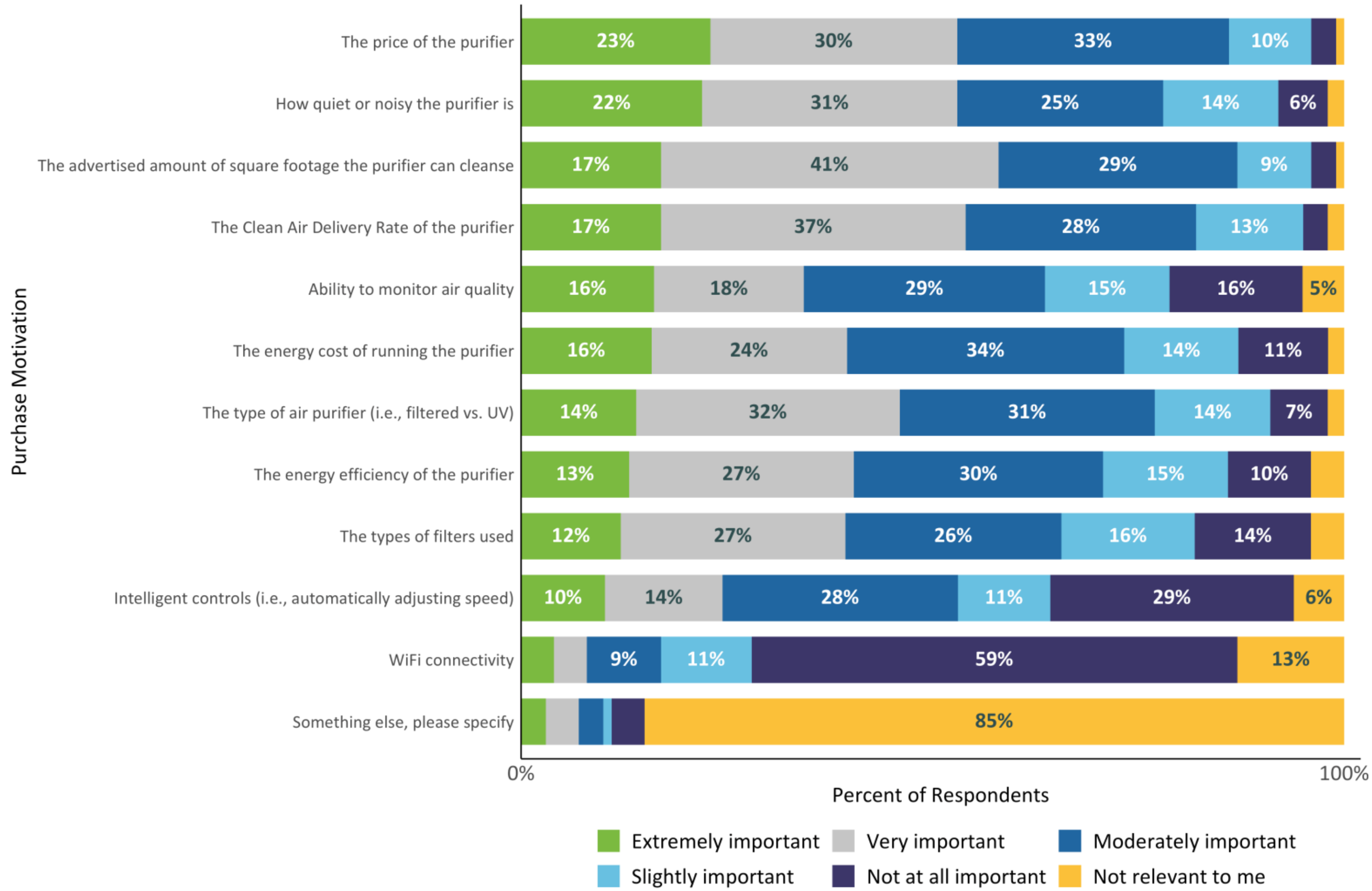
In addition to asking respondents the importance of different drivers that factor into their purchase of air purifiers, we asked which aspects of the purifiers themselves were important when making the purchase. These questions were asked only of respondents who reported being involved in purchasing their home's purifier(s) and were asked at the respondent level instead of at the purifier level.

Overall, the price of the purifier was reported to be the most important aspect when shopping for purifiers, but few factors had an overwhelmingly strong impact on purchasing decisions across the whole sample (i.e., more than 50%; Figure 27). More than 50 percent of the sample rated the following features as very or extremely important:

- The advertised amount of square footage that the purifier can cleanse;
- Clean air delivery rate;
- Unit price; and
- The amount of noise produced by the purifier.

While the above features had the highest rated importance, it seems that most of the factors were important to most respondents to some degree, with the notable exception of Wi-Fi connectivity; most respondents (59%) reported that Wi-Fi connectivity played no role in which purifier they chose. Intelligent controls also proved to be a less popular feature, with only 24 percent of the sample reporting that automatically adjusting fan speed was important to them. These results point to a focus towards bottom-line interests regarding functionality and price, with less of a concern for additional features.

Figure 27: Overall Importance of Air Purifier Features (n=408)

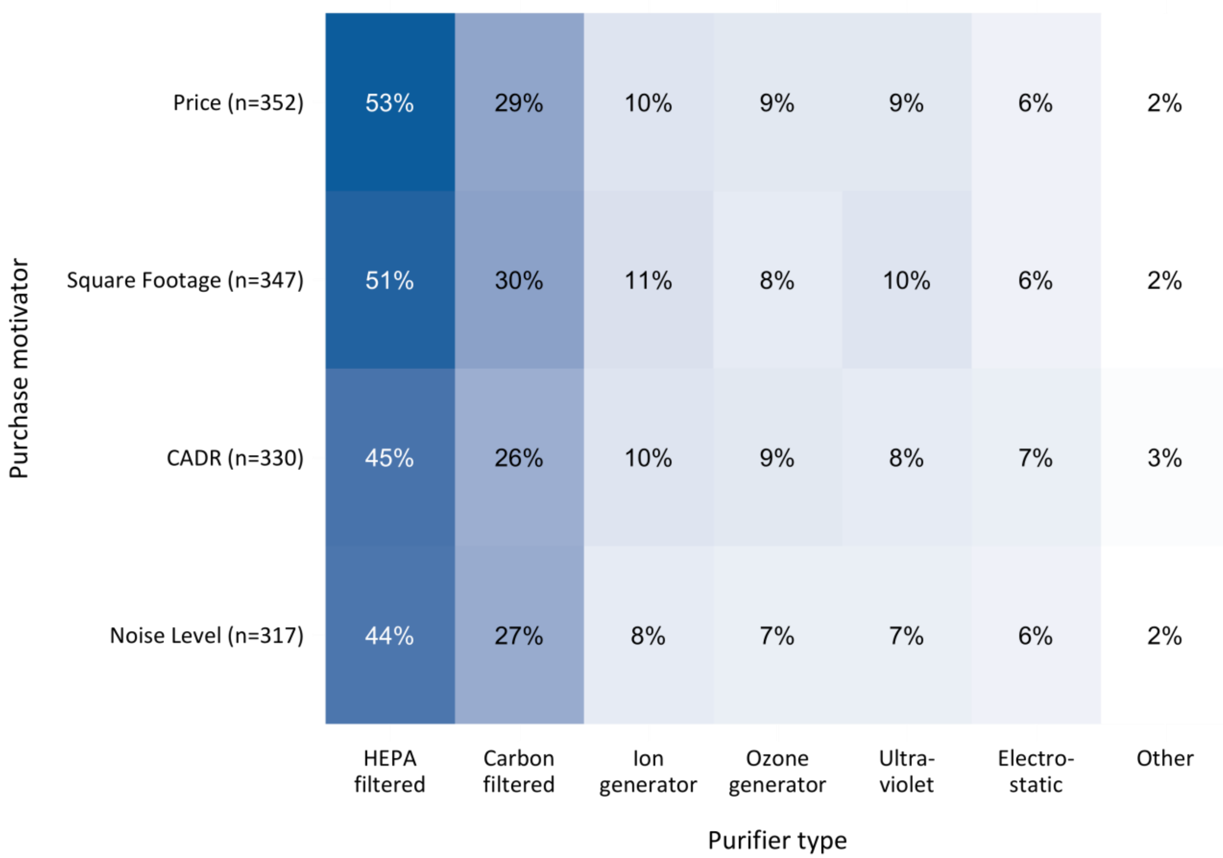


Feature Preferences and Purifier Types

To determine the extent to which the aforementioned purchasing motivations translate into specific decionmaking about which purifier to buy, we examined the distribution of purifier types for those who described different motivators as more or less important.

Specifically, Figure 26 below shows the distribution of purifier types among those who rated the most common preferred features (i.e., price, square footage, CADR, and noise level) as important to them to some degree (i.e., extremely, very, or moderately important). This analysis is presented on the *household* level, rather than the purifier level. For example, 53percent of households that self-reported price as an important feature also reported owning at least one HEPA filter. The distribution of purifier types is roughly the same across each feature, with HEPA accounting for the majority each time.

Figure 28: Purifier Types by Most Important Features



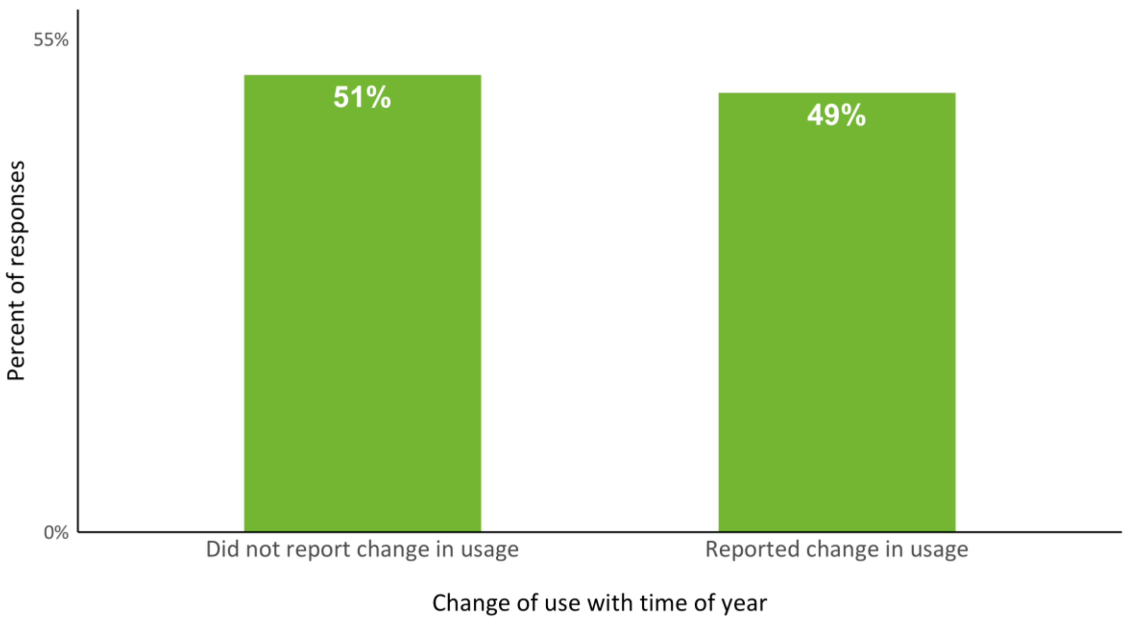
Air Purifier Usage and Habits

Usage by Time of Year

Finally, we asked respondents the extent to which the usage of their air purifiers differed by time of year. If respondents reported that their usage differed by time of year, we asked them to then report how their usage differs by quarter. Note that this question was asked of respondents across all reported purifiers, and thus results are presented here on the respondent level.

Figure 29 summarizes how respondents reported on whether their usage changes by time of year. The sample was roughly evenly split in their estimates, with slightly more than half of the sample reporting that their usage does not change depending on the time of year.

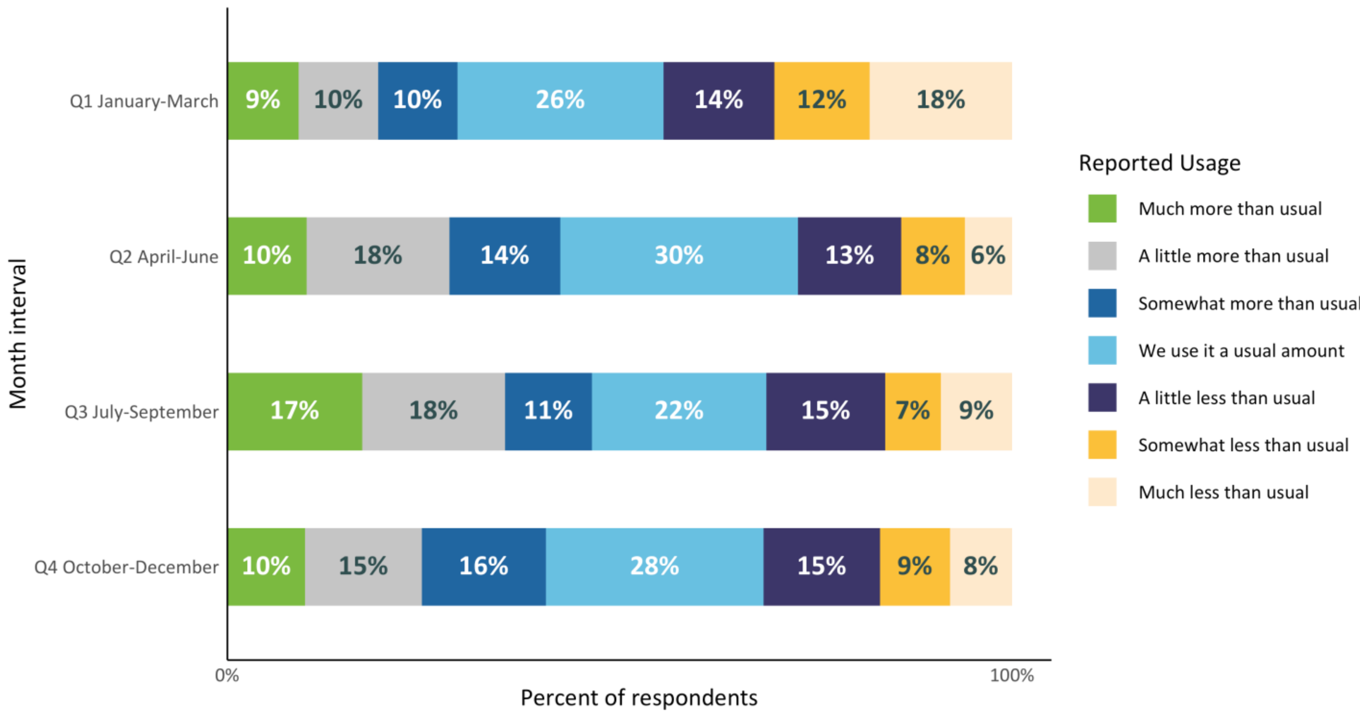
Figure 29: Self-Reported Difference in Usage (n=488)



Respondents did not show clear trends in usage across quarters, whether using their purifiers more or less than usual. For the second quarter (April to December), 41 to 46 percent of respondents who reported differences in use reported using their purifier(s) more than usual, a range of 27 to 32 percent reported less usage than usual, and the rest reported a usual amount of usage. The first quarter period (January to March) provides a possible example of strong usage change; 44 percent of respondents reported using their purifier less during these months

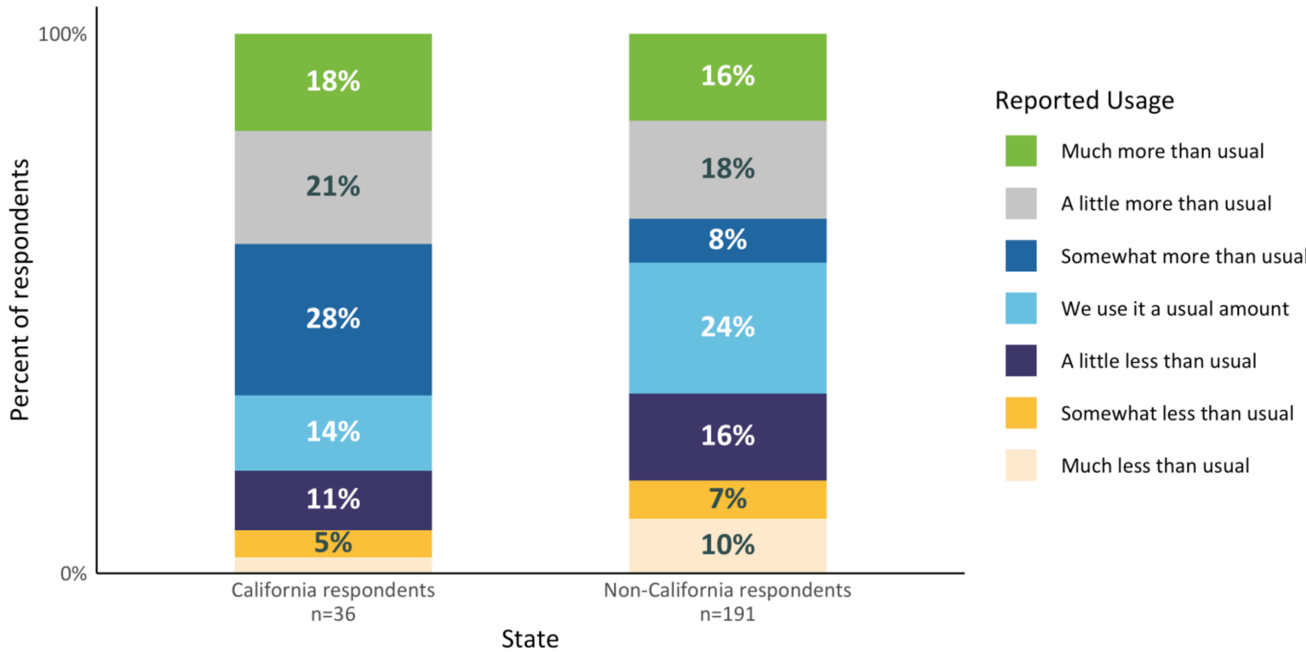
(Figure 30). This finding points to the idea that usage of purifiers is not necessarily strongly impacted by seasonality, despite allergies being a strong driver of adoption. These data seem to point to a trend of year-round usage, with a possible dip at the beginning of the year.

Figure 30: Changes in Air Purifier Usage by Quarter (n=230)



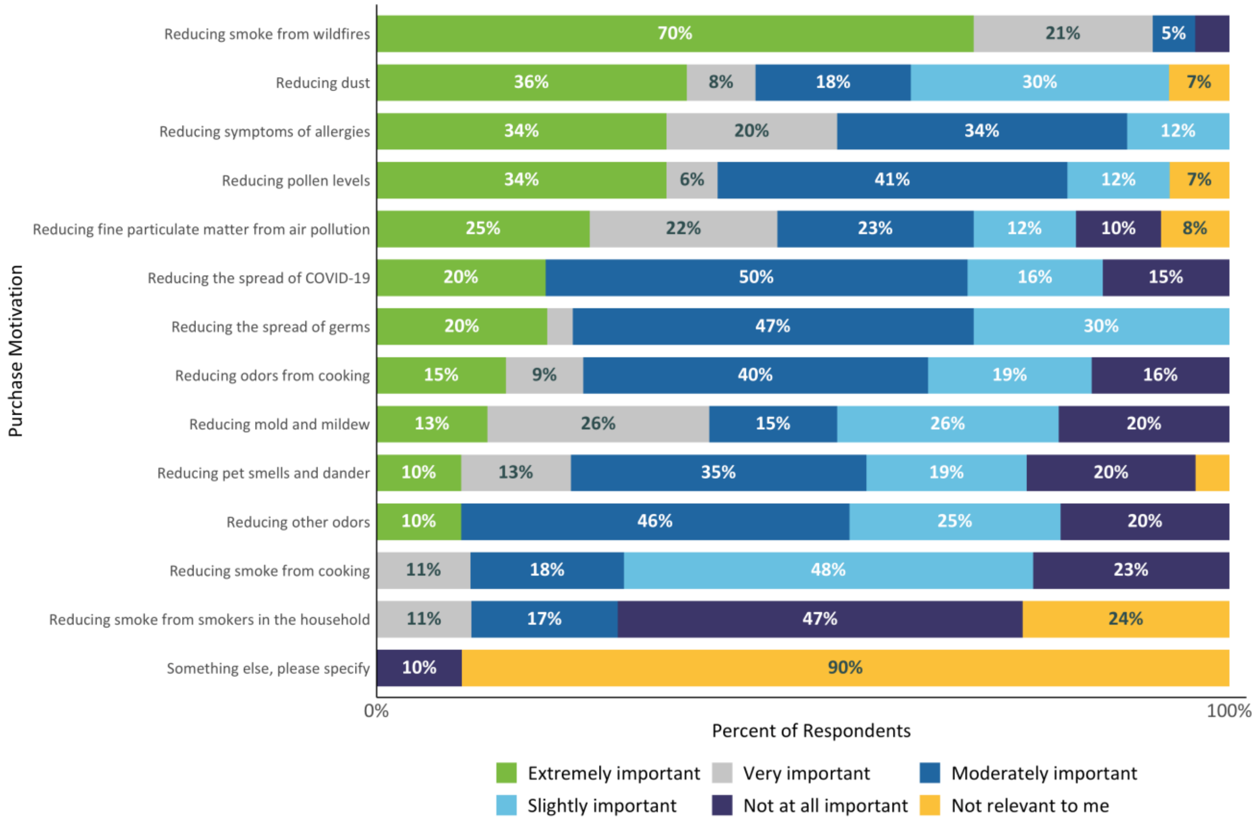
Given the increased usage during the summer months (July – September), we investigated whether this trend persisted in California when compared to non-California respondents. Figure 31 depicts reported air purifier usage in the summer months by California versus non-California respondents. California residents reported a slight increase in usage (i.e., “much more” or “a little more than usual” usage) than non-California residents, though this comparison is not statistically significant.

Figure 31: California Vs. Non-California Summer Usage (n=227)



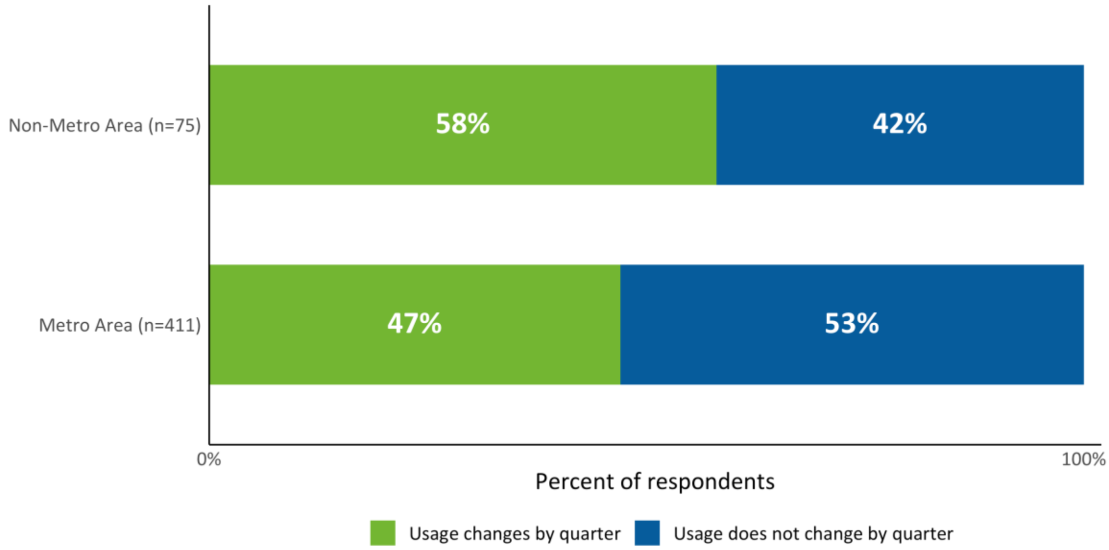
We more closely examined the California respondents who reported much more or a little more than usual usage during the summer months, with an eye towards differences in their motivations for purchasing an air purifier (Figure 32). Notably, the primary purchase motivation for this subgroup is reducing smoke from wildfires, with 70 percent of respondents noting this as extremely important. This finding, while notable, is reported with a very small sample size (n=14).

Figure 32: Purchase Motives of California Respondents with High Summer Usage (n=14)



Respondents in metro areas were 11 percent less likely to report that their usage changes over time, compared to respondents in non-metro areas. Respondents in non-metro areas appear more likely to report changes in usage by quarter compared to the overall sample (58 percent of non-metro respondents versus 47 percent of the overall sample; Figure 33).

Figure 33: Self-Reported Differences in Usage, Metro versus Non-Metro

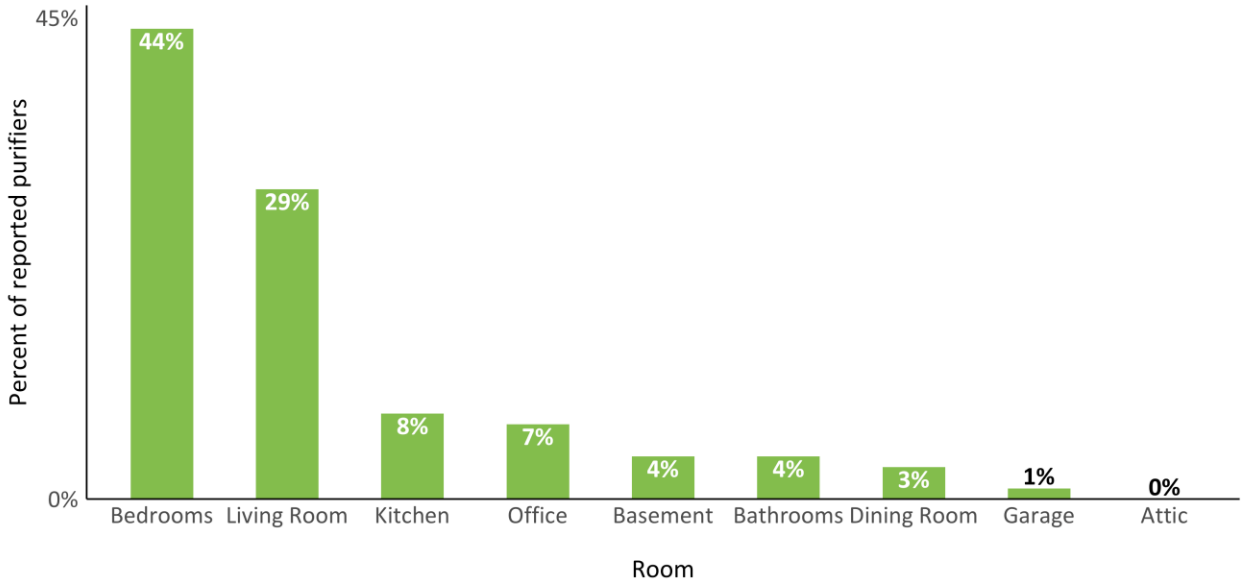


Rooms Where Purifiers Are Used

In addition to asking when they use their purifiers most often, respondents were asked which rooms they use their air purifiers in the most (Figure 34). These questions were asked about every purifier that respondents reported, so results in this section will be reported on the purifier level.

The most reported rooms where respondents run their air purifiers were bedrooms (44%), followed by living rooms (29%). Only 8 percent of reported purifiers were reported as being run in a kitchen, while only 9 percent were reported as being run in office spaces. Less than 5 percent of purifiers were reported as being used in other rooms including basements, dining rooms, bathrooms, garages, and attics.

Figure 34: Distribution of Rooms Where Air Purifiers Are Operated (n=491)



Respondents reported that most air purifiers are run in a single room (76%), followed by 14 percent of purifiers that are run in two rooms and less than 5 percent that are run in three to seven different rooms (Figure 35). Air purifiers that can purify the entire house accounted for 6 percent of air purifiers. The average number of rooms in which air purifiers are run is 1.2 rooms.

Figure 35: Distribution of the Number of Rooms in Which Air Purifiers Are Run (n=489)

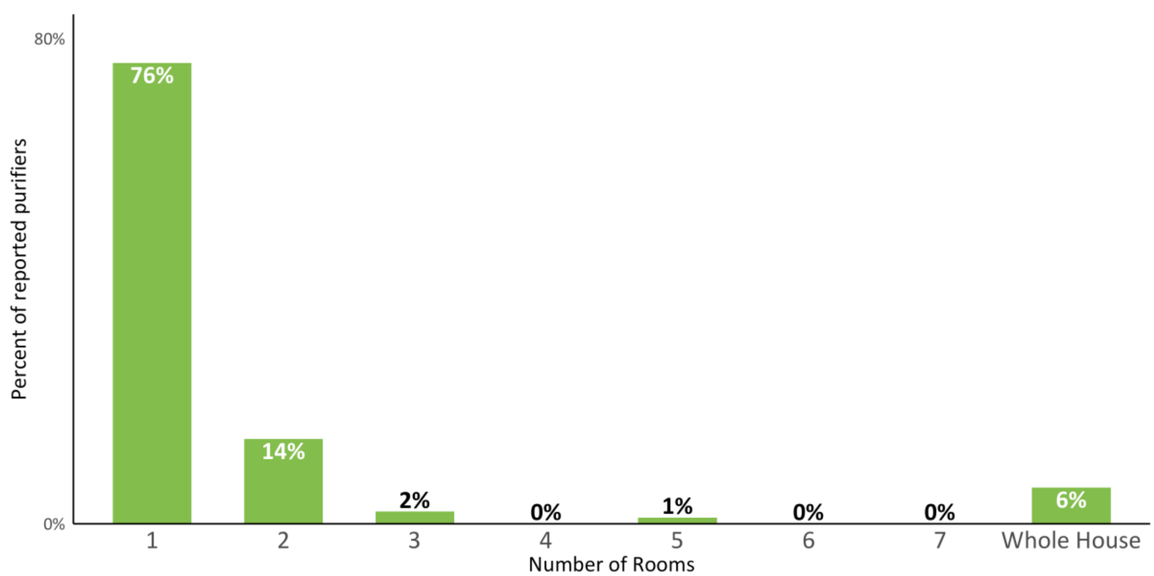
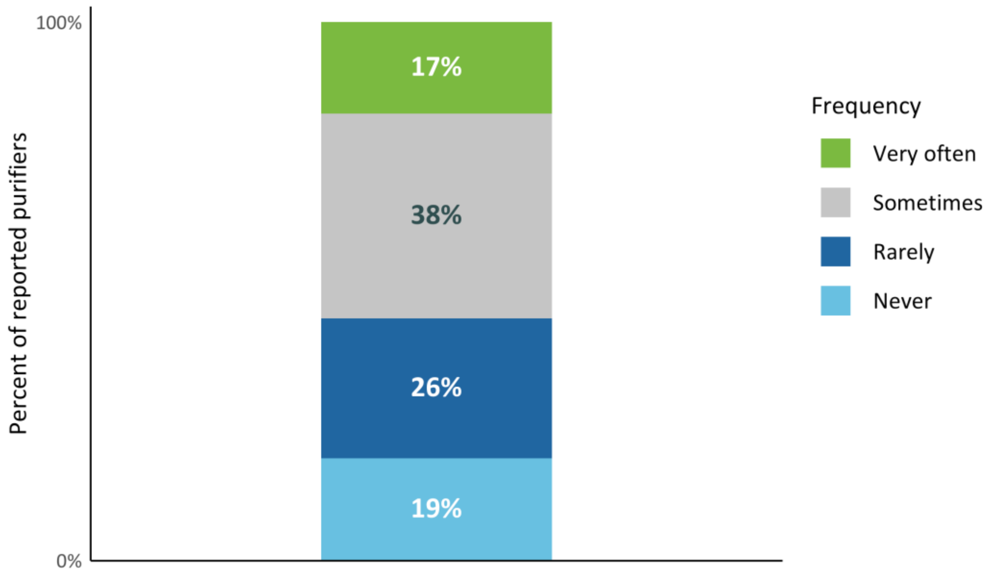


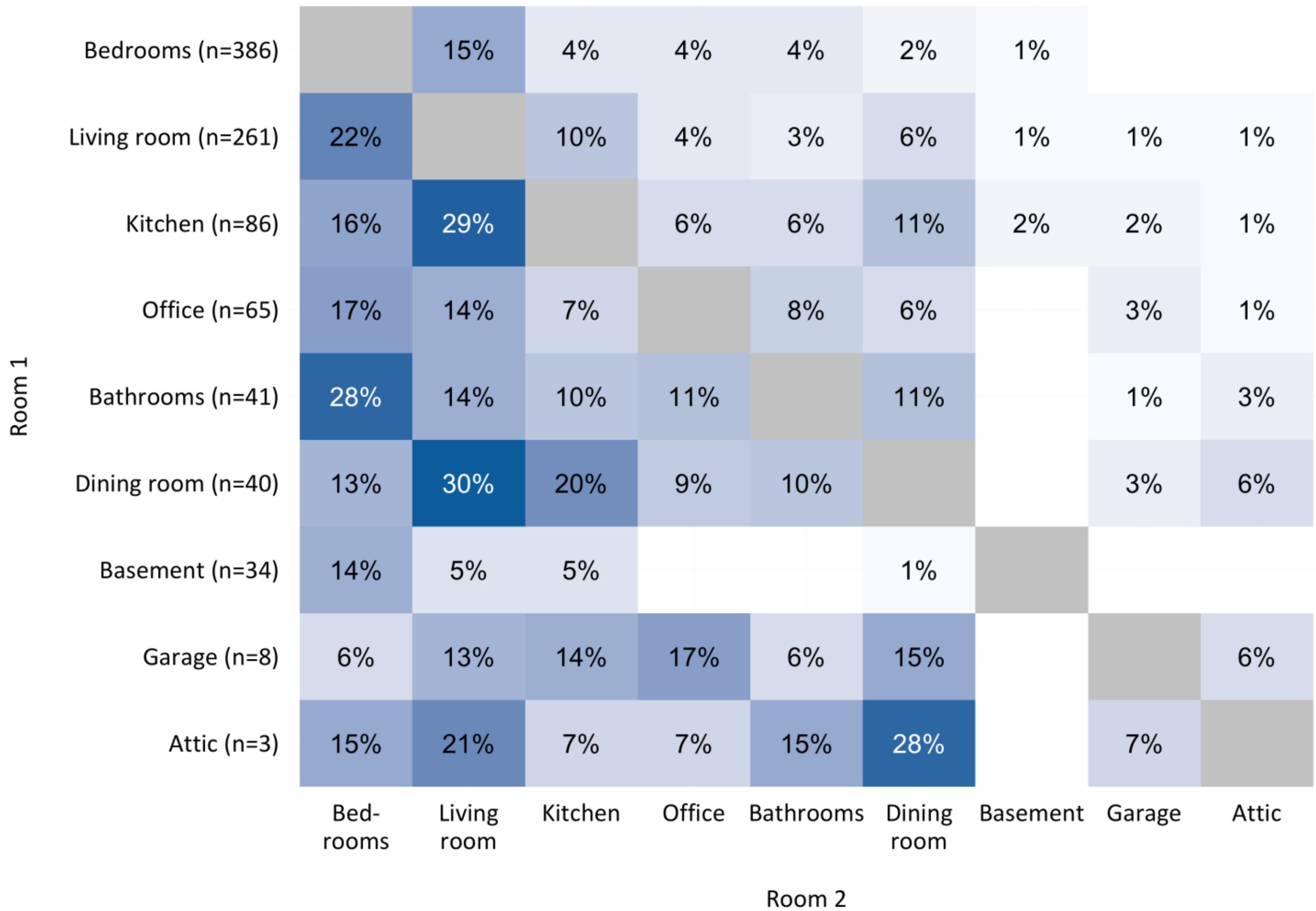
Figure 36 suggests that for the most part, air purifiers are at least occasionally moved from room to room, as only 19 percent of reported air purifiers were described as moved from one room to another. Most reported purifiers (55%) were described as moved to a different room at least occasionally, with the largest proportion of purifiers “sometimes” moved to a different room.

Figure 36: Reported Frequency of Moving Air Purifier Room (n=137)



The most common combinations of rooms in which air purifiers are operated are bedrooms and living rooms (30% of responses). Purifiers that are run in kitchens tend to also be run in living rooms and bedrooms, while purifiers run in dining rooms tend also to be run in living rooms and kitchens. Bedroom purifiers, on the other hand, seem to be for the most part dedicated to bedrooms exclusively, with use in both bedrooms and other rooms relatively infrequent (Figure 37). These results point to a possible distinction in usage for purifiers that are used in common spaces in the home, which may be moved to different locations, versus those in personal or private spaces (such as bedrooms), which are less likely to be moved.

Figure 37: Overlap of Rooms in Which Purifiers Are Used

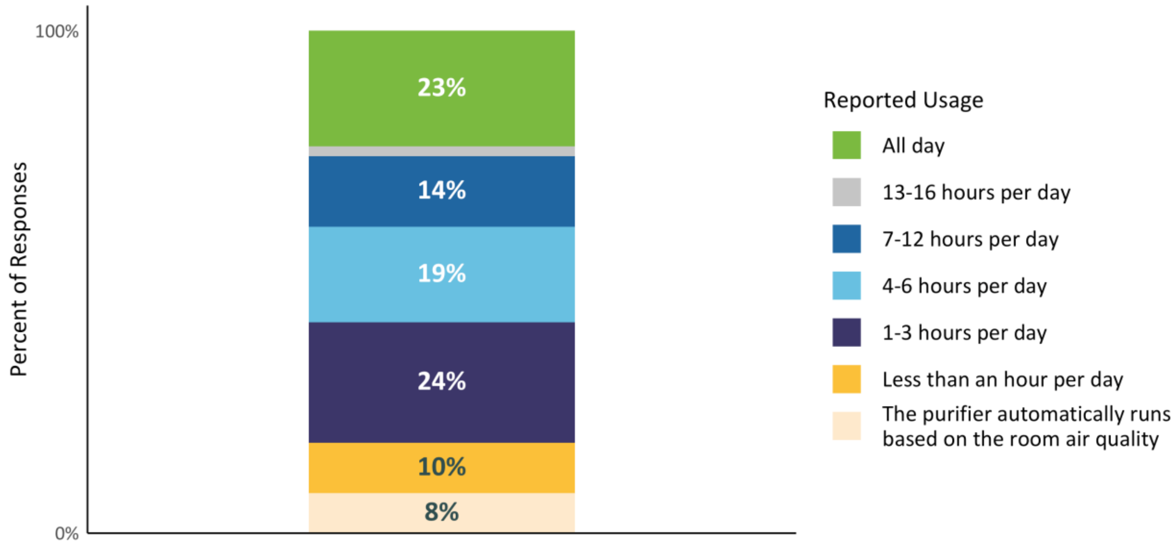


Purifier Usage Time

Finally, our survey evaluated the amount of time that respondents spend running their purifiers daily. These questions were asked across all reported purifiers, and results are shown at a respondent level rather than at a purifier level.

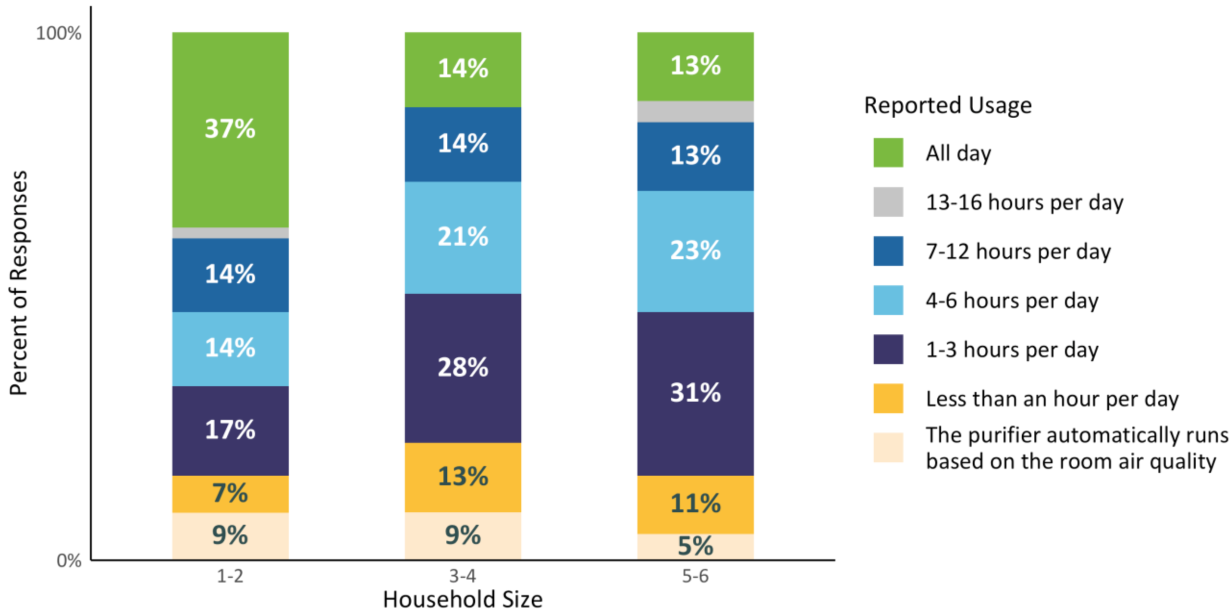
There is widespread usage in terms of how many hours respondents spend running their purifiers, but relatively few (39%) reported running any/all their purifiers for more than six hours a day, with the bulk of those respondents comprised of those who run their purifier all day (23% of respondents; Figure 38).

Figure 38: Reported Usage Frequency (n=488)



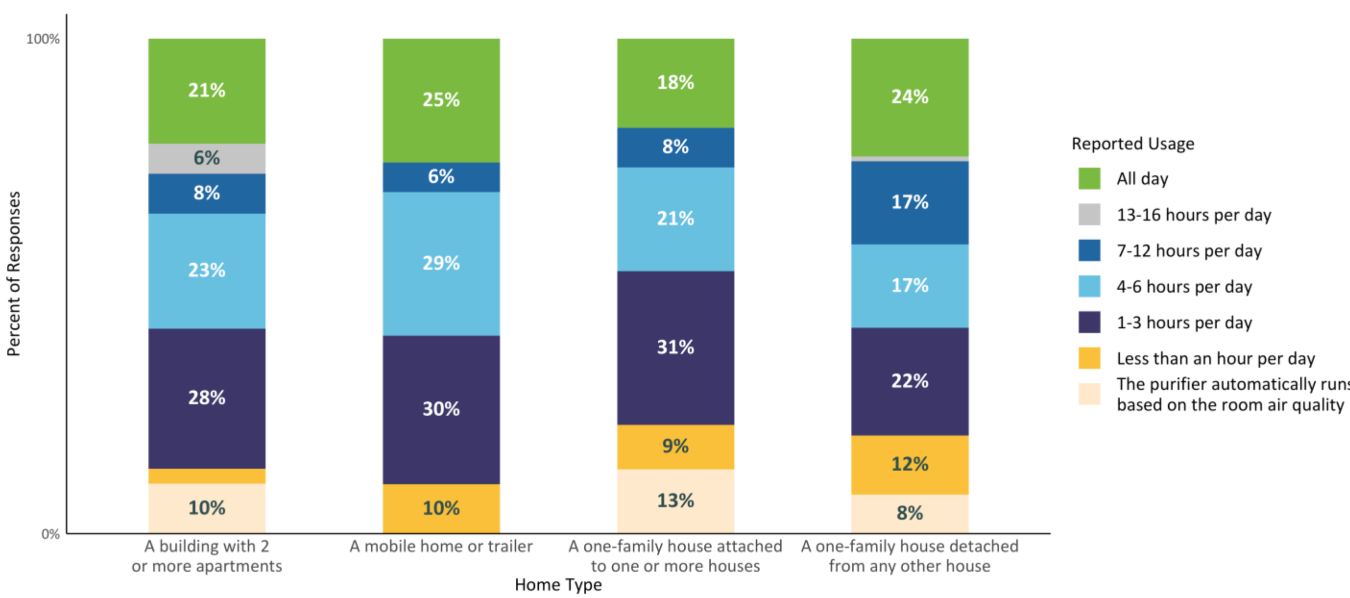
Daily reported usage frequency differs significantly by household size. Households with one or two members reported that 37 percent of their air purifiers are run all day, a much greater percentage than what was reported by larger households (14 percent for households with three or four members and 13 percent for households with five or six members). This difference is statistically significant at 95 percent confidence.

Figure 39: Reported Usage Frequency by Household Size (n=488)



Air purifier usage is generally similar across respondents with different home types (Figure 40). Respondents with mobile homes or trailers are more likely to run their air purifier all day than in any other home type (25% versus 18-24%).

Figure 40: Reported Usage By Home Type (n=488)



Air purifiers owned by respondents residing in the western United States are run in a similar fashion to air purifiers owned by respondents in other regions, with roughly half of air purifiers being run for less than six hours a day for both the western and non-western regions (52% and 53%, respectively; Figure 41).

Figure 41: Reported Usage Frequency By Region (n=488)